Implementation Matters: External Validity of Treatment Effects in Education

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Abstract

Targeted instruction is one of the most effective educational interventions in low- and middle-income countries, yet reported impacts vary by an order of magnitude. We study this variation via meta-analyses of randomized trials across contexts and use the results to inform a new randomized trial and evidence aggregation model. We find two factors explain most of the heterogeneity in effects across settings: accounting for the rate of implementation (intention-to-treat or treatment-on-the-treated effects) and program delivery model (teachers or volunteers). There is debate on whether randomized trial results in one setting, which have high internal validity, are also externally valid in new settings; we find high external validity when considering implementation. Yet, formal accounting for implementation in the literature remains rare: less than 12% of impact evaluations currently account for quantitative metrics of implementation. Our study motivates more frequent estimation and reporting of treatment-on-the-treated effects to enhance external validity and generalizability. Given the importance of implementation, we also introduce a new Bayesian framework to formally incorporate implementation uncertainty and various dimensions of implementation into evidence aggregation. Results show targeted instruction delivers average learning gains of 0.43 SD when taken up and 0.85 SD when implemented with high fidelity.

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1 Introduction

617 million young people worldwide are in school but unable to read fluently or perform simple numerical operations. Although many policies to address learning gaps have yielded disappointing results, a growing evidence base identifies effective educational programs to improve learning (Murnane and Ganimian, 2016). Programs which target educational instruction have shown particular promise in a variety of contexts (Banerjee et al. 2007; Duflo, Dupas and Kremer 2011; Banerjee et al. 2017; Carlana and La Ferrara, 2025; Nickow, Oreopoulos and Quan, 2020).[1] A recent flagship report highlighted targeted instruction as one of the most cost-effective approaches to address the global learning crisis (Global Education Evidence Advisory Panel 2020).[2] However, effect sizes can range from 0.07 to 0.78 standard deviations – an order of magnitude difference.[3] Systematic analysis of this variation could reveal important information on how generalizable effects might be and uncover factors that yield the largest frontier effects in the literature. This is especially important as targeted instruction is adapted and scaled across contexts, with multiple ambitious scale-ups underway. Moreover, to ensure scientifically robust findings, we need to understand how evidence translates across diverse settings and delivery models (Gertler, Galiani, and Romero, 2018; Maniadis, Tufano, and List 2017).

In this paper, we assess the external validity of targeted instruction treatment effects across settings by aggregating evidence across randomized trials. We then use the results to inform a new randomized trial as well as develop a new model for evidence aggregation which formally accounts for various implementation dimensions. For our aggregation, we consider data across 8 study arms covering nearly 75,000 students. We collect data on effect sizes as well as contextual covariates such as baseline learning, geographical context, sample size, year, and implementation delivery model (teachers or external volunteers). We also consider novel data on program implementation first using the notion of "takeup", measured by attendance or presence of classroom materials. Second, we consider the "fidelity" of implementation, measured by adherence to core program principles, such as whether instruction is targeted and students are grouped by learning level as expected. Targeted instruction is an ideal setting in which to study the role of these two aspects of implementation, as both vary widely across studies: takeup ranges from 8% to 90% and fidelity from 23% to 83%.

Given heterogeneity in both program features and reported effects, careful attention to evidence aggregation methodology is required. We provide results from both standard Frequentist random-effects meta-analysis and a series of Bayesian hierarchical models, including meta-regression models which formally incorporate data on program features. We also report several metrics typically used to assses external valdity and generalizability: first, the frequentist I^2 metric. The I^2 reports the heterogeneity in treatment effects as a percentage of the total variation in observed estimates, which is the sum of the genuine variation and the sampling variation.

[1] Targeted instruction groups students in classrooms and tailors instruction to each students' learning level rather than average expected levels determined by a one-size-fits-all grade-level curriculum. A specific model of this approach called "Teaching at the Right Level" was pioneered by Pratham, a large education NGO in India. Another popular approach to target instruction includes tutoring.

[2] Multiple reviews identify targeted instruction as an effective educational approach (Kremer, Brannen and Glennerster 2013; Angrist et al. 2020).

[3] It is important to note that these reported effects are all substantial in a context where a 0.10 standard deviation effect size is considered large (Kraft 2020; Evans and Yuan 2020).

A low I^2 indicates that most of the observed variation in estimated effects was sampling variation rather than genuine treatment effect heterogeneity, indicating greater external validity across settings. We also report the Bayesian standard-deviation in true effects which captures heterogeneity directly, and the Bayesian posterior predictive distribution which captures uncertainty about the predicted effect in the next hypothetical study setting. Given a sample of 8 study arms, which is large for evidence synthesis, but small for frequentist statistical approaches relying on large-sample properties, we prefer the Bayesian approach for evidence aggregation.

While many factors might explain effect size differences across contexts, such as geography or baseline learning, results show that most of the heterogeneity in reported effects is explained by implementation factors, including both formally accounting for the rate of implementation (intention-to-treat vs. treatment-on-the-treated effects) and implementation delivery model (teachers or volunteers). Our Frequentist meta-analysis finds that average intention-to-treat (ITT) effects for teachers are moderate (0.07 SDs) and consistent across studies ($I^2 = 0.01\%$), whereas volunteers have large average effects (0.24 SDs) with high variation ($I^2 = 95.6\%$). When accounting for implementation using instrumental variables estimation to calculate treatment-on-the-treated (TOT) effects, effects for teachers are three times larger (0.21 SD) and consistent ($I^2 = 0.00\%$). Similarly, effects for volunteers are three times larger, with an average effect of 0.76 SDs. Most strikingly, effects now converge almost fully, with TOT effects highly consistent ($I^2 = 0.00\%$). This result reveals that much of the original heterogeneity in volunteer estimates was due to variation in implementation. Thus, conditional on implementation factors, the effects of targeted instruction appear large and consistent across studies.[4]

The Bayesian analysis upholds these patterns, although results are somewhat tempered.[5] The TOT effect is still much larger than the ITT on average, particularly for volunteer delivery, and highly consistent across settings. Moreover, individual studies' TOT estimates see large gains in precision due to partial pooling, combining information from low and high implementation settings.[6] Bayesian meta-regression confirms that both implementation take-up and the implementation delivery model are two key factors in predicting variation in effects. This is not obvious ex-ante, with these dimensions playing a more substantial role than other factors which *a priori* could have mattered most, such as students' baseline learning levels. The evidence on the positive impact of targeted instruction is clear even when we impose strong null priors (a form of Ridge regularization), which suggests the patterns in the data are robust and informative. We show that these results are robust to dropping any individual study and show no evidence of publication bias. Overall, these results show that features of program implementation predict the largest effects in the literature as well as the consistency of effects across settings. While we cannot definitively assess if the consistency we find in treatment effects would translate to future settings, the high rate of consistency across geographies and contexts reveals that randomized trial results, often assumed to be internally valid but not necessarily externally valid, can indeed be highly predictive of effects across diverse settings when accounting for implementation.

These results matter in particular if high implementation rates are not a fixed feature of an environment, and if implementation dimensions can be improved. To address this question,

[4]It is not ex-ante clear this convergence would occur, as noted by de Chaisemartin and Deeb (2023).

[5]This is due to accounting for joint uncertainty in effect averages and variation (Meager 2019).

[6]For example, First UP Camps becomes statistically significant by conventional Frequentist standards.

we conduct a new randomized trial optimizing a specific dimension of the implementation of targeted instruction: the degree or intensity of targeted instruction. We conduct this test in the context of an intervention being scaled up in Botswana which is now active in every region of the country. Implementation is a broad concept, and we focus only on a subset of implementation dimensions in this paper. We randomly vary the degree of targeted instruction, with even more targeted instruction achieved via more detailed learning assessments and grouping of students relative to standard implementation. We conducted a randomized A/B test in a subsample of 52 classes in 4 regions. We find that improved targeting of instruction increases the program's impact by up to 0.22 SD. These results highlight that implementation is not a fixed property of a given setting, but is malleable, and can be optimized in the context of a scaling program.

If implementation is not a static feature of the environment and can instead be altered, then it is a random variable that can and should be studied in a manner similar to treatment effects themselves, particularly as the two are likely to be correlated. To account for this we introduce a new Bayesian model which incorporates data on implementation into the meta-analytic process. This allows us to consider uncertainty in implementation, just as we typically consider uncertainty in treatment effects. The Bayesian approach further allows us to jointly model uncertainty in effects and uncertainty in implementation across studies and contexts. We can also consider multiple implementation factors within the same framework, allowing joint inference on the role of both take-up and fidelity. Our key quantity of interest is the treatment effect that might be attainable at different levels of implementation. Our results show that targeted instruction offers 0.43 standard deviation improvements in learning on average when taken up, and around 0.85 standard deviation gains when implemented with high fidelity, consistent with the upper range of effects in the literature.

Our findings contribute to a literature on education in low- and middle- income countries. Improving learning outcomes is difficult, with decades of stagnant learning outcomes, despite increasing enrollment in school (Pritchett 2013; Angrist et al. 2021). Moreover, input-focused interventions which only provide more resources, such as provision of textbooks or computer hardware, have rarely improved learning (Glewwe, Kremer, and Moulin 2009; Kremer, Brannen, and Glennerster 2013; Beuermann et al. 2015; Ganimian and Murnane 2016). In contrast, pedagogy-focused interventions which improve the quality and type of teaching in the classroom have had far greater success in improving learning, such as targeted instruction and structured pedagogy approaches (Banerjee et al. 2007; Piper et al. 2014; Banerjee et al. 2017; Angrist et al. 2020; Eble et al. 2021; Carlana and La Ferrara 2025). Our results are consistent with this emerging view and show that this insight generalizes across heterogeneous contexts. In contrast with evidence aggregations in other sectors, such as microcredit, which show small or null effects (Meager 2019), we find that targeted instruction has large and consistent effects. There is debate on whether randomized trial results can generalize well across settings, with prominent critiques citing high internal validity yet limited external validity of RCTs (Deaton and Cartwright 2018). In our setting, we find that RCT results can translate strikingly well across settings when formally quantifying and accounting for implementation.

We also contribute to the literature on external validity and advance the practice of evidence synthesis. Researchers are increasingly engaging in evidence synthesis across contexts in ways

that go beyond traditional meta-analysis (Card and Krueger 1995; Banerjee et al 2017b, Andrews and Oster, 2019; Bando, Näslund-Hadley, and Gertler 2019; Vivalt 2020; Bandiera et al. 2021; Meager 2022; Lund et al. 2022; de Gendre et al. 2022; Gechter 2023; Kremer et al. 2023). We contribute to this literature along a few dimensions. We show that accounting for implementation through instrumental variable estimation of treatment-on-the-treated effects enables better external validity of effects across contexts. While instrumental variables estimation has been used to demonstrate credible approaches to generate internally valid estimates, we show new and crucial relevance for external validity. TOT estimation has well-developed frameworks (Imbens and Angrist 1994), but remains infrequent in practice. We review 4000 papers in development from 2019 and 2022. All report ITT effects but only 12.1 percent of education RCTs reported TOT effects, LATE estimates, or implementation metrics.[7] Our results show that to achieve greater external validity, reporting and aggregating TOT or LATE estimates should become common practice. While growing attention in the external validity literature has been given to implementer types, such as public vs. private providers (Romero et al. 2020; Usmani et al. 2024; Angrist et al. 2023), less attention has been paid to accounting for implementation take-up and fidelity rates (Angrist and Hull 2023; de Chaisematin and Deeb 2024). We formalize the essential role of implementation in generalizability analysis, and present a Bayesian model incorporating implementation into the evidence aggregation process.

We also advance a growing literature on the science of scale and implementation (Al-Ubaydli et al. 2017; List 2024). This paper quantitatively accounts for implementation and reveals that implementation is the decisive dimension in generalizing results across contexts at scale, motivating a research agenda on the details of effective implementation, consistent with the notion of "the economist as plumber" (Duflo 2017). Given that targeted instruction yields 0.85 SD when delivered with high fidelity – over 5x higher than the typical education intervention which yields 0.10 SD (Evans and Yuan, 2020) – research on better implementing known productive interventions, such as targeted instruction, can be higher return than discovery of new interventions. Our trial in Botswana provides a concrete example of the return to studying implementation.

The results in this paper have significant implications for policy. Targeted instruction has the potential is on track to reach over 60 million children in South Asia and sub-Saharan Africa by 2025 (J-PAL 2022). High-profile scale up examples include Zambia, where the government has already scaled up to over 3,000 schools, Nigeria, where targeted instruction is being delivered in over 5 states; and Botswana where the government has signed a 9-year Memorandum of Understanding to scale-up nationally. This paper identifies factors that mediate the largest and most consistent effects in the literature, informing adaption and scale-up in new contexts.

The rest of the paper is organised as follows. Section 2 describes the intervention and context. Section 3 describes the data; Section 4 explains our evidence aggregation approach with results in Section 5. Section 6 shows a proof of concept for improving implementation dimensions in a new randomized trial in Botswana. In Section 7 we introduce a Bayesian framework and results which include implementation metrics in the evidence aggregation process. Section 8 concludes.

[7] We include papers captured in the 3ie database, the Top 5 economic journals, top-tier general interest journals (Review of Economics and Statistics, Economic Journal, Journal of the European Economic Association, all American Economic Journal AEJ journals), and a top field journal (the Journal of Development Economics).

2 Educational Intervention and Context

Educational enrollments have increased worldwide to above 90% in all regions, yet learning progress has been much more limited. International education agencies have called this phenomenon a "learning crisis" (World Bank 2018). The learning crisis is most pronounced in low- and middle-income countries (Ganimian and Djaker 2022). For example, in Kenya, Tanzania, and Uganda three-quarters of grade 3 students cannot read a basic sentence. In rural India, half of grade 3 students cannot solve a two-digit subtraction problem such as 46 minus 17 (World Bank 2018). The global learning crisis is estimated to cost over 129 billion USD in lost social welfare (UNESCO 2017).

A combination of factors contributes to the learning crisis, including curricula targeted mostly to advanced students, rote learning, and automatic promotion regardless of learning achieved in prior grades (Banerji and Chavan 2016). Many education interventions have focused on providing inputs to improve learning, such as textbooks, computers, cash transfers, reducing class size, or increasing teacher salaries. However, decades of randomized trials show input-focused initiatives rarely improve learning (Kremer, Brannen, and Glennerster 2013; Evans and Popova 2016; Ganimian and Murnane 2016; Angrist et al. 2020).

In contrast, a pedagogical shift – targeting instruction to the level of the child – has been shown in randomized trials to dramatically improve learning across multiple contexts (Banerjee et al. 2017; Duflo, Dupas and Kremer 2011; Duflo, Kiessel, and Lucas 2024). Targeted instruction involves regrouping students by their learning level (e.g., addition, subtraction) rather than using grade-level grouping determined by rigid curricula. Most education systems are organized to teach a one-size-fits-all curriculum by grade. However, there is often substantial heterogeneity in student learning level in each grade, with most students well below grade-level expectations. For example, a teacher's syllabus might prescribe them to teach division to a class of grade 3 students which is the curriculum-level expectation. Yet, if only 10 percent of the class knows division, 90 percent of the class is being left behind. If a child cannot recognize or add numbers, they will not be able to learn division. Targeting instruction involves regrouping students by learning proficiency rather than by grade. Instead of using mass education that reaches a few, this approach uses customized and engaging teaching and learning that is targeted to the learning level of the child.

A specific model of targeted instruction approaches is called "Teaching at the Right Level" (TaRL), developed by Pratham, one of the largest education NGOs in India. Targeted instruction approaches have been shown to consistently improve learning outcomes for children across diverse contexts. But as yet no systematic meta-analysis across studies has been conducted. A systematic analysis could help identify factors which drive heterogeneity and predict the highest frontier effects in the literature. Making progress on this question has significant implications as targeted instruction approaches are actively being adopted by dozens of countries and scaled up to over 60 million children worldwide. The World Bank, USAID, FCDO, governments, and NGOs are all actively engaged in targeted instruction scale-up efforts (see Figure A1).

3 Data for Evidence Aggregation

Studies included. We consider existing clustered randomized controlled trials which primarily focus on in-school classroom-based models in low and middle-income countries and for which the complete microdata is available. We exclude studies of targeted instruction models which rely on technology, such as mindspark software which adapts to the level of the child (Muralidharan, Singh, and Ganimian 2019) as well as low-tech phone-based tutorials (Angrist, Bergman and Matsheng 2022) which we consider a different intervention. We restrict to studies where the original microdata was available to enable detailed implementation and takeup analysis.[8]

Table 1 shows all included studies and highlights sample characteristics for each. The relevant level of observation is the study-treatment-geography. States in India have a population on the scale of most countries, and are highly heterogeneous, so we include states as a geographical unit in India; moreover, the trials in India often stratify or randomize within a state.

The resulting included trials represent eight geography-treatments. The total sample across the studies is over 75,000 students. We start our analysis with studies recognized as consistent with the targeted instruction model by both the original evaluators and implementers.[9] Moreover, we access the microdata for all these studies, allowing us to replicate original results as well as enhance our ability to conduct a comprehensive evidence aggregation exercise capturing study-level covariates and program features. We focus our analysis on randomized controlled trials to ensure we aggregate causal effects.

Table 1: Studies Considered for Evidence Aggregation

Authors	State/Country	Treatment Arm	Delivery	Sample Size
Studies included				
(1)	Maharashtra, India	Balshaki Camps	Volunteer	10000
(2)	Uttar Pradesh, India	First UP Camps	Volunteer	9442
(3)	Kenya	Tracking	Teachers	6000
(4)	Bihar, India	School Volunteers	Volunteer	3325
(4)	Bihar, India	Teacher Camps	Teachers	2474
(4)	Uttar Pradesh, India	UP 10-day Camps	Volunteer	17266
(4)	Uttar Pradesh, India	UP 20-day Camps	Volunteer	13054
(4)	Haryana, India	In-school Teachers	Teachers	11966
Total	5	8	-	73527

Notes: These treatment arms are included in the following studies: (1) Banerjee, Cole, Duflo & Linden (2007); (2) Banerjee, Banerji, Duflo, Glennerster & Khemani (2010); (3) Duflo, Dupas & Kremer (2011); (4) Banerjee, Banerji, Berry, Duflo, Hannan, Mukerji, Shotland, Walton (2017).

[8] If microdata become available, we can include an intervention in Ghana (Duflo, Kiessel, and Lucas 2024).

[9] For example, we exclude some treatment arms in Banerjee et al. (2017) which only provided generic materials, but did not provided targeted instruction support. We also exclude some treatments where targeted instruction did not occur such as in Uttarakhand as verified in carefully collected monitoring data.

Mitigating concerns around publication bias. We explore the potential for publication bias in the targeted instruction literature and the possible impact on our findings. Figure 1 shows the distribution of t-statistics for treatment effects. Previous work has observed that publication bias is probable if we observe a jump in t-statistics right above the 1.96 cutoff which is a conventional threshold for statistical significance (Brodeur et al 2016, Andrews and Kasy 2019). We do not observe such a jump, and rather observe more studies right under this threshold as well as t-statistics which are much larger. A possible reason for this distribution is that the sample sizes in this literature are very large, limiting potential for manipulation of significance thresholds.

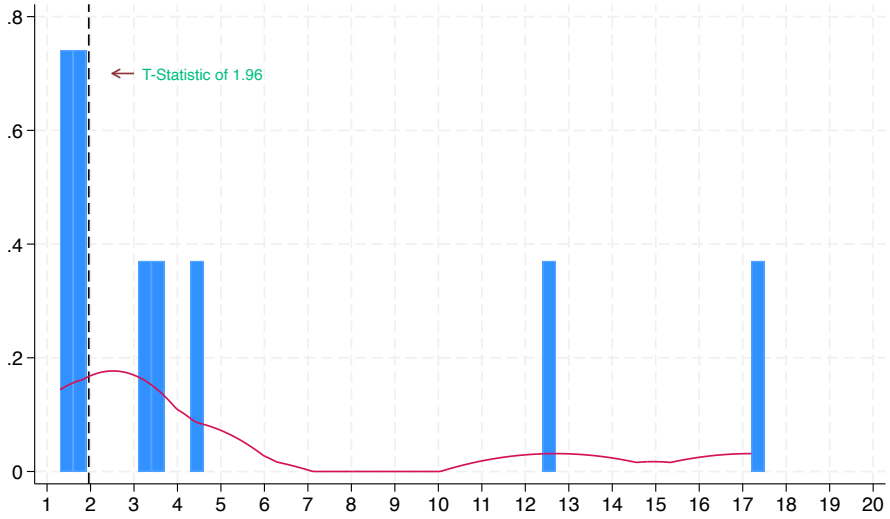


Figure 1: Distribution of t-statistics in our sample

Notes: This figure shows a histogram and kdensity plot of t-statistics.

Outcome Data and Measurement. We access the microdata from each study to produce new standardized outcome variables. In all studies, the central outcome is a measure of learning basic numeracy and literacy skills. Most studies use an assessment similar to the Annual Status of Education Report (ASER) test. Figure A2 shows examples of ASER assessments for literacy and numeracy. Our primary outcome is an average of numeracy and literacy.

The ASER test is a validated learning measure which tests competencies used across 14 countries and is consistently used in the education literature (Banerjee et al. 2017). In numeracy, questions include number recognition, addition, subtraction, multiplication, and division. In literacy, competencies tested include letter recognition, word recognition, ability to read a sentence fluently, and reading comprehension of a paragraph and short story. The Kenya study is the only study which uses a different assessment which is a 100-point test which also covers basic numeracy and literacy. We examine average scores over both numeracy and literacy. In most cases both subjects are available, however in two cases only one subject is available. All studies include baseline and endline data and some studies also include midline data. For consistency and to capture longer-lasting effects, we focus on effects at endline. To compare these outcomes,

we standardize each score relative to the standard deviation within a state/country-treatment unit. We derive learning gains over the course of one year to compare outcomes on a consistent time horizon. Given our underlying assessments measure a similar outcome and most use a similar test, comparability of outcomes across contexts is high.

One of the advantages of conducting a meta-analysis of targeted instruction approaches is that the outcomes are similar across intervention settings. In many cases, meta-analyses rely on outcomes which can vary substantially, derived using entirely different surveys and definitions of outcomes, such as in the case of microcredit (Meager 2019, Vivalt 2020, Pritchett and Sandefur 2015). Our relatively uniform outcome data is well suited to aggregation and offers a substantial improvement in the comparability of treatment effects across contexts. We analyze effects in terms of standard deviation (SD) units, which is a widespread practice in the education literature. Results are also robust to using raw units.

Implementation data. The original studies report intention-to-treat (ITT) effects using random assignment to estimate treatment effects. We replicate and calculate new harmonized intention-to-treat effects across studies.

We calculate novel estimates of treatment-on-the-treated (TOT) effects for those who actually received the program using instrumental variables estimation in line with the Local Average Treatment Effects (LATE) framework (Imbens and Angrist 1994). In some cases TOT effects were originally reported, but not in most cases, so we calculate new estimates for all studies.[10] Formally accounting for implementation via take-up rates is likely to be central to understanding both the average impact and the generalizability of the evidence, since in some studies take-up is over 80 percent while in other studies it is around 10 percent. It is noteworthy that in many cases implementation was very high. This reveals that while implementation can vary, it can also reach near-complete levels, increasing the relevance of understanding effects for those who take-up the intervention, representing the majority of the population of interest in most settings. In Appendix Table A1 we also show that the subset of compliers who implement are largely representative of the broader sample along a set of key characteristics such as gender, grade, age, and baseline level of learning.

Implementation is a broad concept – we focus on two few specific implementation dimensions: take-up and fidelity. We define take-up as the share of the population that received the intervention. We define implementation fidelity as the degree or intensity of implementation, with a focus on how targeted instruction is likely to be.

We include data captured via random monitoring visits to schools to collect process monitoring information: teacher attendance, student attendance, and whether the expected materials were present in the classroom. We calculate the share of schools visited where each indicator was verified. We consider take-up as an average across these indicators. We interpret this measure as the share of the population that received some intervention. We include this measure in our Frequentist meta-analysis since it is available for all studies. The last measure goes beyond take-up to capture implementation fidelity – how targeted the instruction was based on student

[10]Thus we require access to the underlying microdata from the original studies in order to quantify the degree of implementation and fit new aggregation models to these variables. We have take-up data for all studies, and we have fidelity data for a subset of studies.

grouping – available in three of the intervention arms. This is based on data from random monitoring visits on whether students appeared to be grouped by learning level. We incorporate this measure into our Bayesian evidence aggregation in later sections, since Bayesian methods are best suited to small-sample inference.

Additional data and covariates. We standardize and incorporate a series of additional data and covariates likely to mediate effects across studies. These include: geography (country or state), implementation delivery model (teacher or volunteer), year of intervention, sample size, and baseline learning levels.

Replication. We replicate original results prior to conducting a new meta-analysis. We find broadly consistent results with original reported estimates. No signs change and findings remain robust. Average differences across all studies are less than 0.1 standard deviations. In a few rare cases, the magnitude of estimates differ from original estimates, but only slightly. Reasons for this variation include our use of endline assessments across the board[11], our uniform approach to constructing standard deviations, and our primary measure being an average of scores across both subjects. In addition, in our replication we do not include control variables for consistency since controls varied across studies, which has minor effects on final estimates.

4 Methodological Approach

4.1 Estimating ITT and TOT Effects For Each Study

To facilitate evidence aggregation, we estimate treatment effects for each study using a uniform approach as described in the data section. Moreover, we produce new estimates of treatment-on-the-treated effects (TOT) across all studies in addition to intention-to-treat (ITT) effects. While original studies calculated ITT effects, TOT effects were not always calculated, and we provide novel estimates of treatment-on-the-treated effects across all studies using instrumental variables analysis.

We start by estimating intention-to-treat (ITT) effects of targeted instruction on learning using a consistent estimation procedure across studies. We calculate effects for each study, exploiting random assignment to identify causal effects of being randomized to targeted instruction treatments as follows:

$$Y_{ij} = \alpha + \beta_1 TargetedInstruction_j + \epsilon_{ij} \tag{4.1}$$

where Y_{ij} is a learning outcome for individual i in school j. $TargetedInstruction_j$ refers to any of the targeted instruction interventions and takes on the value 1 in the treatment arms and 0 otherwise.

We next explore the extent to which results are driven by the degree of implementation, focusing on take-up which is a variable we have across all studies, and calculate treatment-on-the-treated effects. Since those who are actually treated might have been treated due to non-random reasons, such as teacher motivation to implement in areas with lower learning levels

[11] The original studies reported a midline assessment for one study.

to begin with, we cannot simply analyze effects for those who received the program. Rather, we use instrumental variables estimation of causal effects, instrumenting for implementation of targeted instruction with random assignment to receive the program, following the Local Average Treatment Effects (LATE) framework (Imbens and Angrist 1994). While the assumptions for instrumental variables estimation to identify causal treatment effects can prove challenging to verify in quasi-experimental research designs, they often hold in the case of randomized controlled trials, the study design for all studies included in our analysis.[12]

A first criterion for reliable instrumental variable estimation is a strong first stage – that is, whether randomization indeed leads to implementation. We indeed find a strong first stage as expected given implementation of targeted instruction is a direct function of random assignment to the program with large F-statistics, indicating a highly statistically significant relationship (as shown in Table A2). A series of additional assumptions need to hold for instrumental variables estimates to capture causal effects. A monotonicity assumption ensures that everyone who would participate when not offered the program would also participate when offered it. This assumption is satisfied by design of the treatment assignment since those not randomized to receive targeted instruction cannot access it. A final assumption is the exclusion restriction, where randomization to targeted instruction only affects learning outcomes through the channel of receiving targeted instruction. The most common reason this assumption might fail is if allocation of the treatment might also alter overall resourcing for the school, which in turn could influence learning through other channels. In the case of targeted instruction programs, this assumption, however, is rarely violated since targeting instruction operates not by adding substantial resources to a school (e.g., there are no general school grants available nor a longer school day) but rather by redirecting an existing inefficient hour of the school day to be more productive through a very specific pedagogical intervention. Thus, the exclusion restriction is likely satisfied since randomization to the program has limited benefits or channel for influence beyond receiving the program itself.

We estimate the causal effect of receiving the targeted instruction program on Y_{ij} as follows:

$$Y_{ij} = \alpha + \beta_1 I_j + \epsilon_{ij} \tag{4.2}$$

where I is our measure of implementation. We instrument for I using random assignment to receive the targeted instruction program. As described in the data section, implementation of targeted instruction is first captured as take-up. Later, using the Bayesian evidence aggregation model accounting for implementation we also consider fidelity captured by whether students were grouped by their learning level.

In our data, implementation rates can be as high as 90%, revealing that local average treatment effects identified do not apply only to a small group receiving the program in practice, but rather can be relevant for a substantial share of the population. Appendix Table A1 further shows that the subsample of compliers who implement are similar to the broader sample along a set of key characteristics, such as baseline level of learning.

[12]For example, while a national policy to reform education might be rolled out inconsistently and bundled alongside other interventions, randomized trials for a particular program, such as targeted instruction, are direct, focused, and often result in a clearly identified treatment.

4.2 Evidence Synthesis – Aggregating Effects Across Studies

We next systematically aggregate effects across studies. Using meta-analysis, we analyze variation in treatment effects of targeted instruction across studies and explore external validity of treatment effects across settings. We use the term external validity to refer to whether a treatment effect in one setting is predictive of a treatment effect in another. Treatment effects can predict one another if they are consistent with each other or if there is a consistent extrapolation function. Generalization and external validity are related concepts, with external validity referring to extrapolation between specific settings, while generalizability refers to extrapolation from a given setting to a broader setting. While we cannot definitively assess if the consistency we assess in treatment effects in our study sample would translate to future settings, consistency of treatment effects in a sample of diverse geographies and contexts is highly informative and a common metric used to plausibly assess external validity and generalizability.

We conduct various types of meta-analyses. Aggregating evidence from different settings requires joint estimation of average effects and heterogeneity in effects across studies. The statistical challenge is to separate genuine heterogeneity in effects from sampling variation and simultaneously use this variation to inform the uncertainty on the average impact. Hierarchical models are able to perform this decomposition (Gelman et al 2004; Meager 2019). However, the interdependent uncertainty between the means and the variances creates a challenging joint inference problem, particularly with a small number of studies. In this setting, Bayesian methods can offer improved tractability and estimation performance relative to popular frequentist counterparts such as random effects or Empirical Bayes (Rubin 1981; Gelman et al. 2004; Gelman and Hill 2007; Chung et al. 2013; Chung et al. 2015).[13] We use a set of Bayesian hierarchical models which estimate the average treatment effect across all studies and the variance across contexts in line with Rubin (1981), Gelman et al (2004), Vivalt (2020), Bandiera et al. (2021) and the Cochrane Handbook version 5.1 section 16.8. Appendix Section C includes more details on the Bayesian hierarchical modeling approach.

An advantage of meta-analyses is quantitative synthesis of evidence. Moreover, systematic synthesis generates statistics which can be used to gauge average effects sizes across contexts, such as the I-squared statistic. The I-squared metric measures the percentage of total variation which is genuine variation in treatment effects rather than sampling variation. A low I-squared indicates that most heterogeneity is due to sampling variation, rather than true treatment effect variation.[14] We also report the Bayesian hypervariance which measures the variation in effects, and the posterior predictive distribution which captures uncertainty about the predicted effect in the next hypothetical study setting.

In some disciplines meta-analyses are seen as the next critical step to follow after conducting individual randomized controlled trials, enabling systematic aggregation of internally valid studies across studies and contexts to inform both internal and external validity. However, others argue that meta-analyses are atheoretical and often compute average effects sizes without

[13]The Bayesian hierarchical framework also permits multiple comparisons, automatically adjusting for multiple testing problems since marginalization of the joint posterior appropriately conditions on all the evidence available in the sample and priors.

[14]If the sample size in each study was small, a low I-squared could be due to noisy estimates. However, when the sample sizes of the set of studies analyzed is large, as it is in our set of studies with around 75,000 observations in the microdata, the I-squared is unlikely to be mostly influenced by noise.

creating coherent classes of interventions to aggregate (e.g. averaging effects of interventions that are quite different).

In this paper, we aim to draw on the benefits of systematic and quantitative study aggregation, while also ensuring we aggregate coherent classes of interventions and delivery models, and inform our meta-analysis with theory and qualitative expertise. Bayesian synthesis lends itself particularly well to this approach, especially since one can capture expertise and theoretical insight through informed choices of priors. We use both frequentist random-effects synthesis, a typical meta-analysis approach, and Bayesian hierarchical models, which is our preferred approach.

5 Evidence Aggregation Results

5.1 Frequentist Random-Effects Results

We start with a Frequentist random effects aggregation, contextualizing the Bayesian aggregation results by providing results without any formal incorporation of theory or priors. First, we aggregate the evidence on the Intention to Treat (ITT) effects, shown in Figure 2. We find that intention to treat effects for interventions delivered by teachers have an average effect of 0.07 standard deviations. These effects are consistent with an I-squared of 0.01 percent, suggesting any variation between estimates is sampling variation rather than true heterogeneity. This implies the teacher delivery of targeted instruction are highly consistent across the programs in our data set. Second, we observe that volunteer delivery is on average three times as effective as teacher delivery with a 0.24 standard deviation effect. However, the volunteer results are highly heterogeneous with an I-squared of 95.6 percent.

Second, we aggregate the evidence on the Treatment on Treated (TOT) Effects that we constructed from the microdata. Results are shown in Figure 3. We observe two trends. First both teachers and volunteers are three times as effective conditional on implementation with 0.21 and 0.76 standard deviation average effects, respectively. Both effects are large and precisely estimated. Most strikingly, we now observe convergence among volunteer effects, with an I-squared of zero. This reveals that much of the heterogeneity in the original volunteer estimates was due to variation in implementation. These results indicate the initial degree of generalizability of targeted instruction in our setting. Effects are large and consistent when accounting for rates of implementation (ITT or TOT effects) and when considering implementation delivery model (teachers or volunteers). The high consistency in treatment effects after accounting for these factors leaves little room for a role for other features. We confirm this finding formally via metaregression.[15]

[15]It is also worth noting that the patterns in this analysis show few diminishing returns to program scale, suggesting that effects might persist as the program is scaled up: some of the largest effects, such as those in the Uttar Pradesh 10 and 20-day camps, have the largest sample size (with up to 17,000 students).

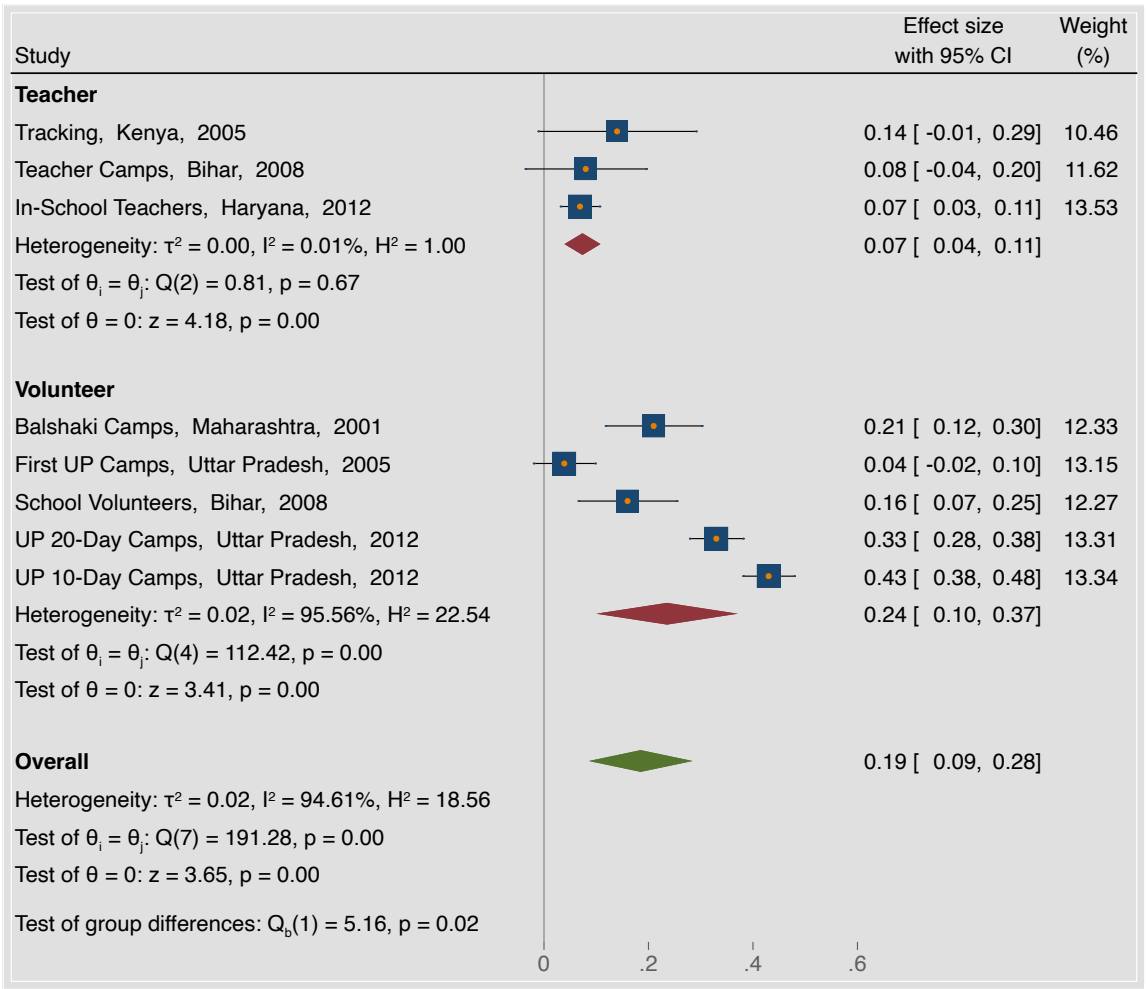


Figure 2: Frequentist Random Effects Meta-Analysis of Intention-to-Treat Effects.

Notes: This figure shows the results of a random-effects Frequentist meta-analysis of ITT effects. Each row represents a study setting and treatment effect and includes the confidence interval as well as the relative weighting in the final average estimates. The figure also reports I^2 statistics which capture the degree of heterogeneity that is due to true treatment effect variation across studies relative to sampling variation.

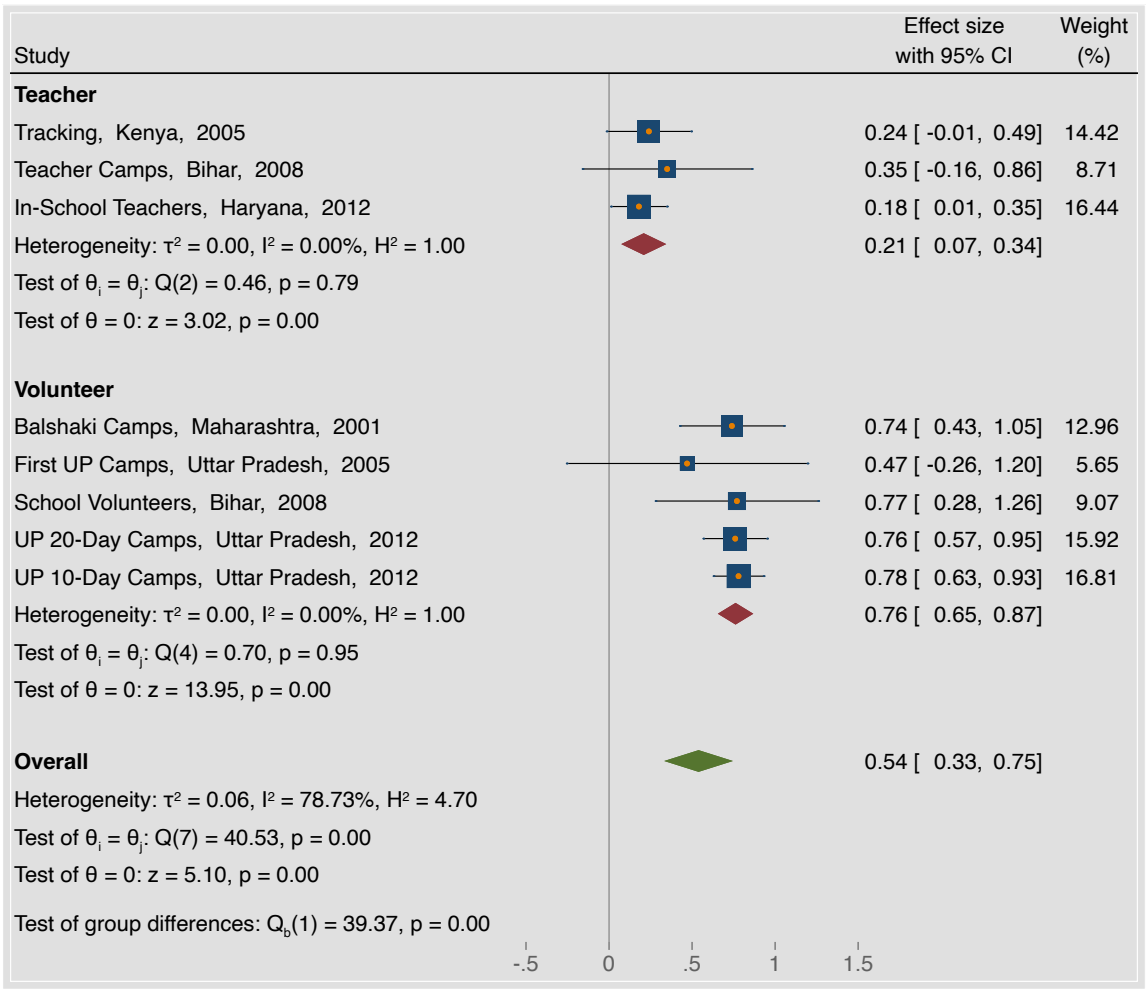


Figure 3: Frequentist Random Effects Meta-analysis of Treatment-on-Treated effects

Notes: This figure shows the results of a random-effects Frequentist meta-analysis of TOT effects. Each row represents a study setting and treatment effect and includes the confidence interval as well as the relative weighting in the final average estimates. The figure also reports I^2 statistics which capture the degree of heterogeneity that is due to true treatment effect variation across studies relative to sampling variation.

5.2 Bayesian Aggregation Results

We now present the results of the Bayesian evidence synthesis. These models correspond conceptually to the Frequentist random-effects model but with joint estimation of the variance in effects and the mean over all the studies rather than sequentially (e.g., partial pooling), and with the potential to incorporate various choices of priors.

We investigate the role of implementation delivery model (volunteer or teacher) fitting the Rubin (1981) model to each subset – teachers versus volunteers – separately. We start by incorporating wide priors centered at zero to regularize the estimation given the limited number of studies available in such aggregation exercises. We compare results using partial pooling in our Bayesian aggregation directly to the no pooling case to understand the extent of information pooling across contexts. In Appendix Section E we present posterior treatment effects for expected results in future settings, as shown in Figure A4.

We present ITT results split by delivery model in Figure 4 and TOT results split by delivery model in Figure 5. The findings show the importance of the delivery model, especially for the TOT results where we see even greater pooling due to the substantial similarity in effects. The visual clustering suggests that when we account for implementation rates (TOT vs ITT effects) and implementation delivery model (teachers vs. volunteers) the treatment effects are highly consistent. The broad patterns found in the frequentist analysis are confirmed in these two figures: the TOT effects are much larger than the ITT effects on average, and also more homogenous. However, there are several interesting differences to note. ITT estimates are relatively unchanged when pooled using Bayesian aggregation; there is slightly more pooling but it remains negligible overall. This is due both to the relative precision of the ITT estimates and their heterogeneity across settings. Further confirming this, in Appendix Tables A6 and A7, we report Bayesian pooling factors which are the reciprocal of the I-squared metric. The ITT sees very little pooling, so most of the variation is true treatment effect variation.

However, the Bayesian model pools the TOT estimates to a substantial degree as shown in Figure 5 (also shown in Appendix Tables A6 and A7). Correspondingly, the precision of each study's TOT estimate is enhanced significantly. For example, "First UP camps" effects which is positive but not significant in the no-pooling case is now statistically significant under partial pooling. This is likely due to low implementation in this setting (only 8 percent of students attended sessions) so TOT effects are hard to estimate and inherently noisy without pooling. Pooling studies with high implementation settings more precisely captures information about the latent effect when implementated. When average TOT effects are similar across studies, as is true in our case, Bayesian aggregation pools TOT estimates where implementation is low with TOT estimates with high implementation studies, enhancing precision substantially.

These results show that average intention-to-treat effects and the treatment-on-the-treated effects are large and positive. Even accounting for joint uncertainty and using priors that regularize results towards zero under uncertainty, these findings hold. Moreover, Bayesian aggregation confers the advantage of enhancing precision for each individual study, in particular for individual TOT estimates.[16] Finally, consistent with the Frequentist results, effects are generalizable across settings when considering both TOT effects and implementation delivery models.

[16]Meta-regression results in Appendix Section D further further confirm results in this section.



Figure 4: Bayesian aggregation of ITT by implementation delivery model

Notes: This figure shows the results of a Bayesian hierarchical model meta-analysis of ITT effects. Each row represents a study setting and treatment effect and includes the confidence interval as well as the final pooled estimates. The figure reports both no pooling scenarios as well as partial pooling scenarios for direct comparison.

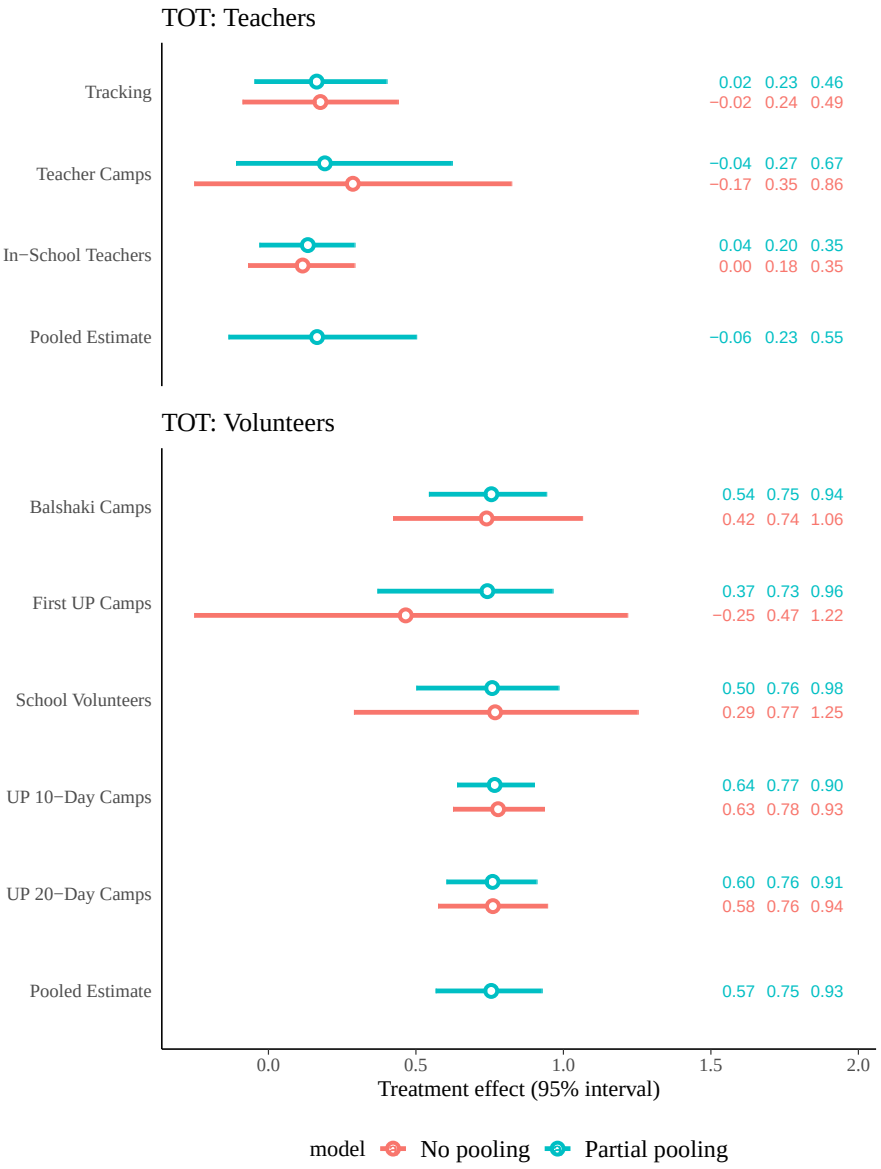


Figure 5: Bayesian aggregation of TOT by implementation delivery model

Notes: This figure shows the results of a Bayesian hierarchical model meta-analysis of TOT effects. Each row represents a study setting and treatment effect and includes the confidence interval as well as the final pooled estimates. The figure reports both no pooling scenarios as well as partial pooling scenarios for direct comparison.

5.3 Results with Informed Priors based on Theory and Expertise

We now combine the findings of the Bayesian statistical analysis in the previous section with qualitative expert insight and economic theory to inform priors. First, we discuss theory. Second, we review the literature. Third, we present results using both types of information - theory and expertise based on the literature - captured formally in the model via priors.

Theory-informed priors. We start by formalizing components of the theory of change. According to an extensive literature as well as qualitative expertise, targeted instruction is designed to bridge learning gaps when student learning levels are far behind grade level. In this environment the curriculum is mismatched to students' zone of proximal development. Moreover, targeted instruction works by creating homogeneous groups which enables more efficient instruction by minimizing the likelihood of mismatch in a given group of receivers of information. This approach is most needed in schooling systems for which in the status quo student learning is far behind grade level and where there is significant heterogeneity and thus high mismatch between curricula and any given student learning level.

The conceptual framework outlined above predicts that students in lower learning levels are most likely to gain from the intervention. This is consistent with a broader economic notion of diminishing marginal returns. On the other hand, there is related economic theory on the notion of complementarities, whereby adding one activity increase the returns of the other. A similar notion conceptually is behind much of the "big push" development literature that underpins many highly influential development programs including the Millennium Villages Project and the BRAC Graduation program, sometimes called "Targeting the Ultra Poor" (Banerjee at al. 2015). This intuition can also apply to the targeted instruction intervention: for example, once a child can recognize numbers, they can more easily learn to do addition, consistent with the prominent notion of dynamic complementarities in skill formation (Cunha and Heckman 2007). This suggests that the students who start at higher baseline levels can progress faster.

Since these theories have qualitative predictions that go in opposite directions, the overall implication for the quantitative estimation model is that one should regularize the correlation between status quo learning levels and treatment effects across settings. This corresponds to a prior that places equal weight on positive and negative correlations but higher likelihood on moderately sized correlations of either sign than on extreme correlations of either sign – a smooth, classical regularization in the style of the Ridge penalty.

Expertise-informed Priors. We now consider a set of stronger priors. We augment the above discussion with additional information based on the literature on educational interventions in developing countries. To ensure that this information is generated prior to any of the evidence on targeted instruction models contained in our present data set, we limit ourselves to literature published before the early 2000s. At that time, there was substantial focus on access to school, which was often assumed to result in learning, and education quality was often proxed by inputs rather than outcomes (Barro and Lee 1996). Indeed, going to school was reasonably expected to be the major production function for learning. Any additional interventions beyond access to school were likely to be marginal in comparison.

An exception to this general pessimism about any single intervention beyond schooling being central to improving learning was surrounding the possibility of providing inputs and incentives to teachers, although deeper discussions in the academic literature noted that there seemed to be a potential role for pedagogical improvement, and that the incentives might primarily work to improve pedagogy, but this potential was largely speculative (Hanushek 1995). Overall, the development economics literature was largely focused on schooling access, inputs, and incentives.

Thus, the state of the field's understanding prior to targeted instruction models further motivates a reasonably tight prior around a zero effect size for this type of intervention. Such a prior encapsulates the qualitative notion that targeted instruction would have to overcome priors against it to prove itself in that intellectual climate. To investigate the results of using strong theory-driven priors, we present results under a variety of strong priors. Using a range of strong priors allows us to understand exactly how strong the patterns in this papers' data and statistical analysis are, and also captures insights from a broader set of information beyond the present studies.

Results with theory- and expertise-informed priors. Figure 6 shows the results of the basic aggregation model for the ITT results under a variety of priors on the average effect, and Figure 7 shows the same analysis for the TOT effects. All priors we consider are Gaussian as discussed in the Appendix and centered at a zero effect, yet their strength varies substantially by varying the standard deviation of the Gaussian prior around zero. For both ITT and TOT estimates, we show the results of a strong pessimistic prior on the effect, represented by a prior SD on the hypermean of 0.5 outcome units, and an even more pessimistic prior represented by a smaller SD of 0.25. These are both a full order of magnitude tighter than the default prior. We also show the extreme case of a near-dogmatic prior, $N(0, 0.1)$, presented as a sanity check (such a strong prior should almost certainly exert a large influence on the posterior).[17] The strong pessimistic priors exert some influence, but only the near-dogmatic prior can promote any noticeable shift. Overall the data shows strong support for positive and substantial ITT and TOT effects across sites, with patterns in the data overcoming strong zero effect priors.[18]

Overall, the patterns in the data are robust and informative about the benefits of targeted instruction, which aligns with the progression of expert opinion. While in the 1990s there was a strong prior on the potential effectiveness of education reforms mostly focused on increased schooling and inputs, after decades of rigorous evidence, an emerging view is that pedagogy reforms, rather than additional inputs, resources, or incentives alone, are the most promising interventions (Global Education Evidence Advisory Panel 2020). This shift in opinion had to overcome strong priors, and our analysis shows that the empirical evidence on targeted instruction is strong enough to help facilitate that shift.

[17] For reference, later we will fit models with a hypermean prior SD of 10, so this is two orders of magnitude tighter than what we consider reasonable.

[18] In Section H.1, we also show results using a model that implements regularization of the correlation between baseline levels and treatment effects. We find similar patterns.

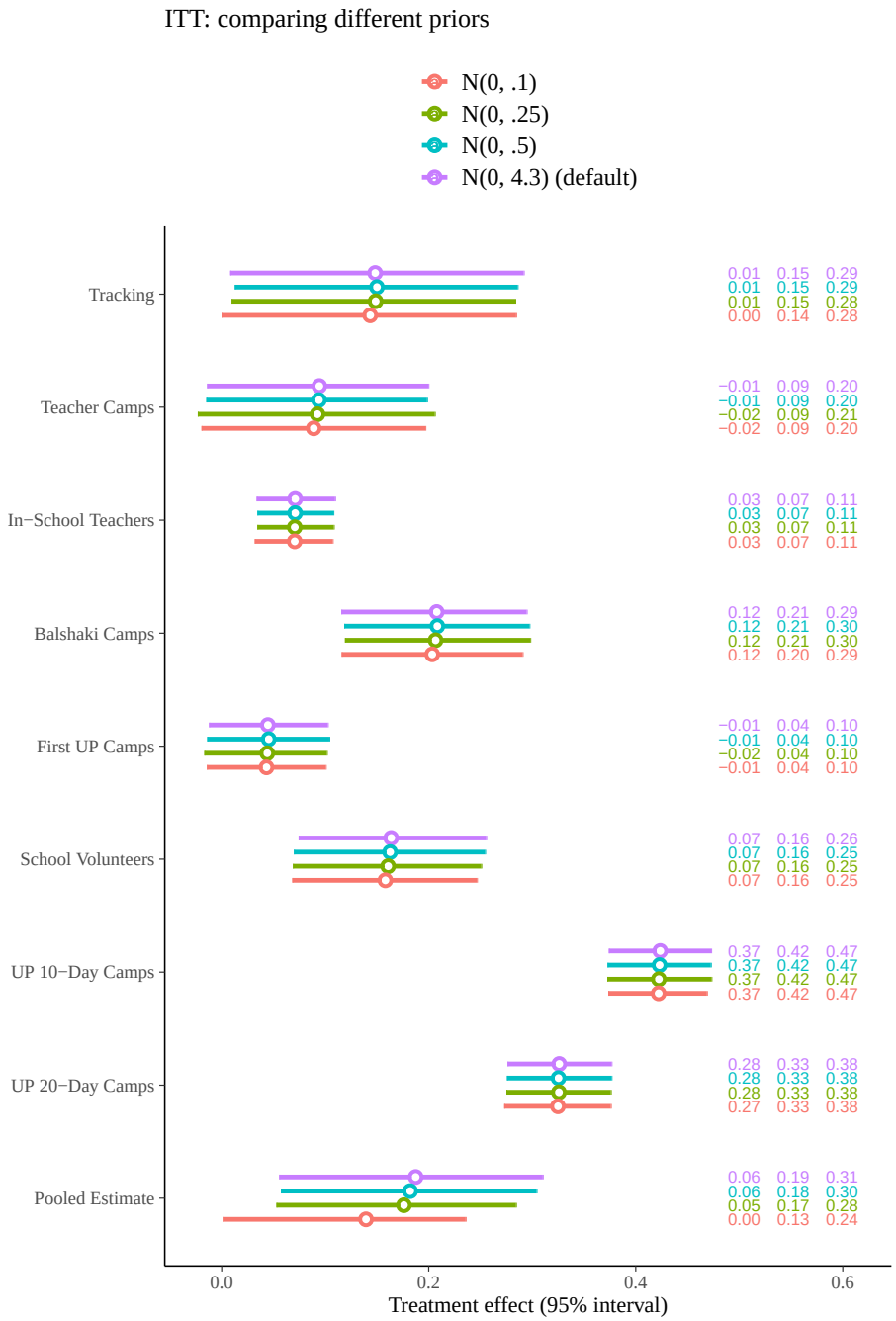


Figure 6: ITT results under different theory-driven priors

Notes: This figure shows the results of a Bayesian hierarchical model meta-analysis of ITT effects with varying strengths of pessimistic priors around zero. Each row represents a study setting and treatment effect and prior and includes the confidence interval as well as the final pooled estimates. We show the results of a strong pessimistic prior on the effect, represented by a prior SD on the hypermean of 0.5 outcome units, and an even more pessimistic prior represented by a smaller SD of 0.25. These are both a full order of magnitude tighter than the default prior. We also show the extreme case of a near-dogmatic prior, $N(0, 0.1)$, presented as a sanity check.

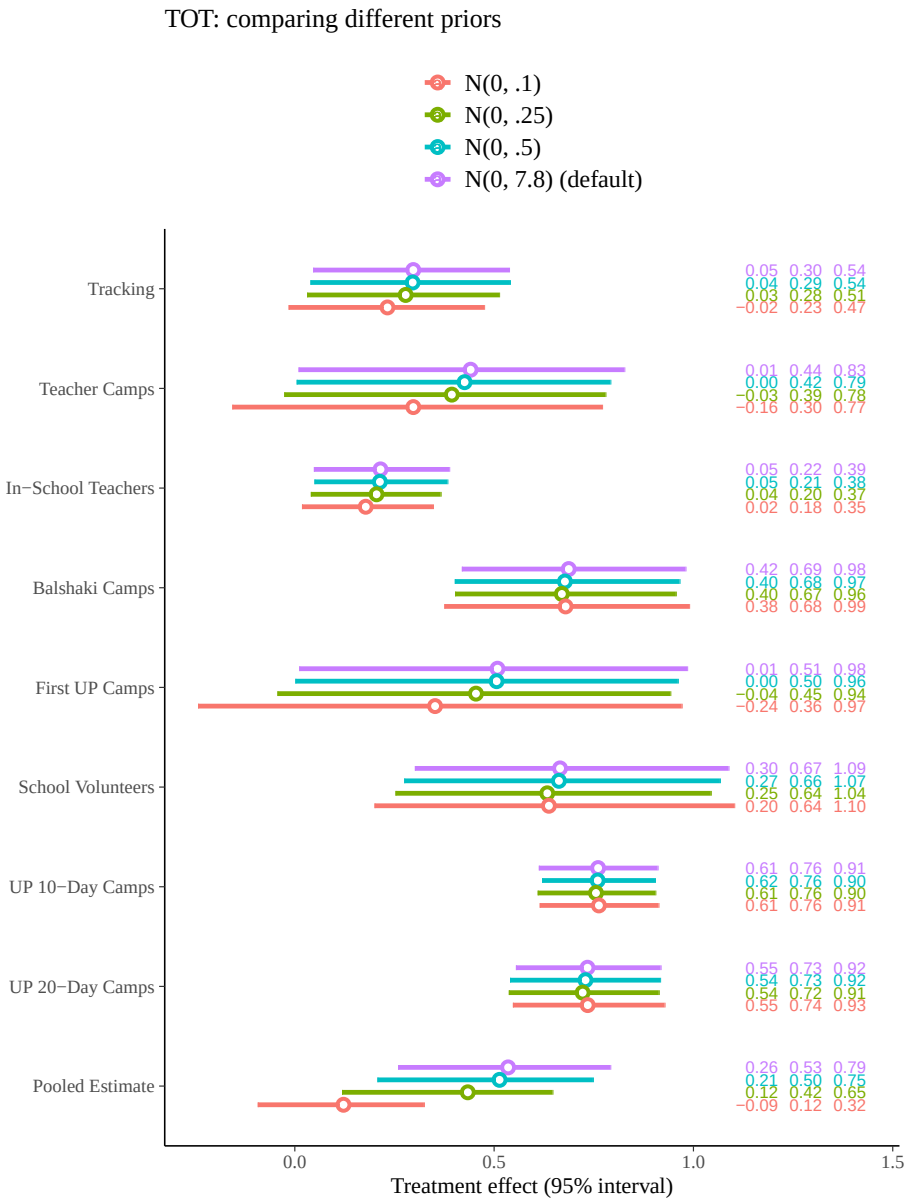


Figure 7: TOT results under different theory-driven priors

Notes: This figure shows the results of a Bayesian hierarchical model meta-analysis of TOT effects with varying strengths of pessimistic priors around zero. Each row represents a study setting and treatment effect and prior and includes the confidence interval as well as the final pooled estimates. We show the results of a strong pessimistic prior on the effect, represented by a prior SD on the hypermean of 0.5 outcome units, and an even more pessimistic prior represented by a smaller SD of 0.25. These are both a full order of magnitude tighter than the default prior. We also show the extreme case of a near-dogmatic prior, $N(0, 0.1)$, presented as a sanity check.

6 Improving Implementation: New Evidence from Botswana

The results of our evidence aggregation establish the importance of implementation in explaining variation in treatment effects across settings. A natural next question is whether implementation can conceivably be changed in practice and in future settings, or if implementation is a fixed feature of an environment. We conduct a new randomized trial to test whether implementation can be improved, and in particular if targeted instruction can be made even more targeted. We empirically test whether there are concrete ways to increase the rate of targeted instruction in the context of a scaling program in Botswana, where the government is actively scaling and testing the program in partnership with Youth Impact, one of the largest NGOs in the country. As of 2022, 20 percent of schools in the country had been reached, with all primary schools expected to be reached by the end of 2026.

6.1 Intervention and Study Design

In Botswana, targeted instruction is implemented by grouping students by operation level during numeracy classes; that is, whether they can add, subtract, multiply, or divide, or do no operations at all (referred to as "beginner"). At baseline in our sample there is a lot of variation and low performance along this dimension. Table 2 below shows the highest operation a child can do at baseline in term 1 of the school year in 2020. The plurality of grade 3-5 students, 30 percent, can do no operations ("beginner" level), 28 percent can do up to addition, 20 percent can do up to subtraction, 15 percent can do up to multiplication and only 6 percent can do up to division. As table 2 shows, however, in this sample of students there is variation along other relevant proficiencies as well, such as the ability to recognise and interpret larger-digit numbers. While 23 percent of students can recognize up to 4 digits, most children cannot, with 45 percent recognizing only up to 3 digits, and 29 percent of students able to recognize up to 2 digits.

Table 2: Learning Levels at Baseline in Botswana

Operations	Proportion of Students
Beginner	0.30
Addition	0.28
Subtraction	0.20
Multiplication	0.15
Division	0.06
Number Recognition	
0 digits	0.00
1 digit	0.03
2 digits	0.29
3 digits	0.45
4 digits	0.23

The selected lever to increase rates of targeted instruction was to increase the likelihood that children receive instruction that is even further targeted to their learning level. Specifically, students were grouped both by operational level *and* digit recognition level. Youth Impact conducted a randomized trial comparing two options to subgroup students. Youth Impact internally refers to this as "A/B testing", conducting regular randomized optimizations every school term.

The standard implementation in schools in Botswana ("Option B" in this trial) involves testing and grouping students according to their understanding of operations and then running operation-specific classrooms (e.g., subtraction class in one room, multiplication in the next). This means that the operation-level classes occur with student groups who have mixed number recognition abilities. For example, subtraction-level students who recognize 3 digits would be in a small group wih subtraction-level 1-digit students. The new treatment being trialed randomly ("Option A") involves additionally subgrouping students within an operations-level class according to their digit recognition level. For example, subtraction-level students who recognize 3 digits would be in a small group together with subtraction-level students who recognise 3 digit, with instruction now further targeted to both operation levels and digit-recognition levels. Figure 8 showcases both versions of the intervention. The trial took place with over 1,069 students across 52 classes in 4 regions in Botswana, randomized at the class level. Table A3 in the Appendix shows baseline balance results, indicating randomization generated equal comparison groups as expected.

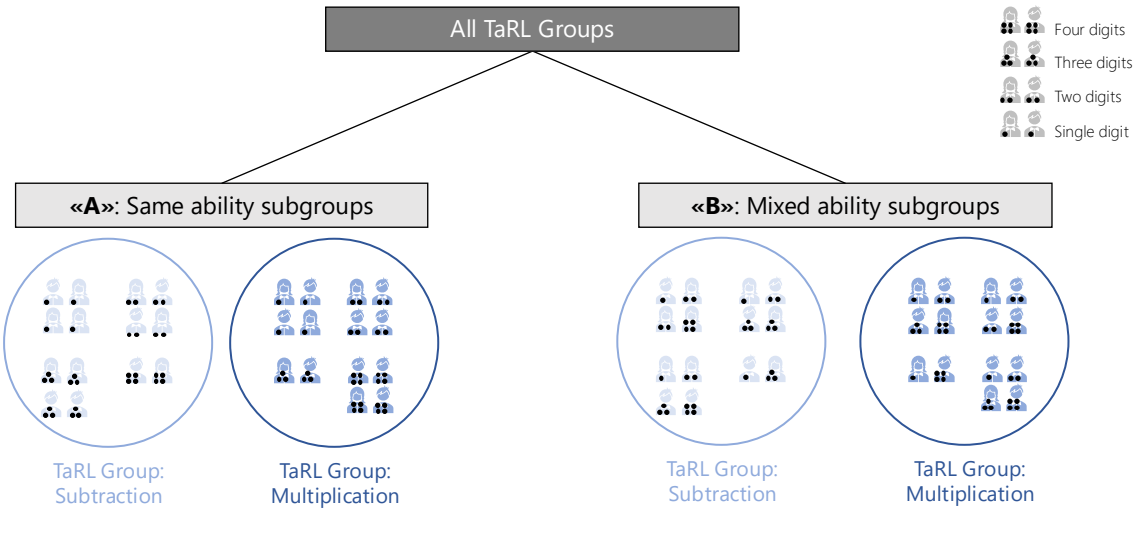


Figure 8: Optimizing implementation: an example further targeting instruction

Notes: This figure shows a visual representation of the randomized study and groups being compared. In Group A, there is grouping based on operation level and digit level, producing even more homogeneous same ability groups, and in turn even higher fidelity to the targeted instruction principle. In Group B, there is grouping on operation only, the default grouping in targeted instruction in Botswana.

6.2 Results

Table 3 reports results of the trial, with the data analyzed using a standard linear regression model estimated via ordinary least squares. Results show that additional sub-learning-level grouping improves learning outcomes by 0.19 to 0.23 standard deviations on average with high statistical significance (p-values range from 0.058 to 0.017). Across columns (1) through (4) we control for multiple characteristics such as region and baseline learning levels. These results reveal that there are sizable gains to further targeting instruction not just along one competency (e.g numeracy operation) but along multiple competencies (e.g. numeracy operation and digit recognition). The effects of this trial are considered large in the education literature where successful programs have effect sizes typically around 0.10 standard deviations (Evans and Yuan 2020). These results demonstrate that implementation is not a fixed of an environment, nor a black-box; rather, improving implementation can be rigorously studied, concrete, tractable, and high-return.

The marginal cost of the targeted instruction optimization in this trial is small, estimated at just a few cents. As a result, the cost-effectiveness of optimizing targeted instruction ranks among the most cost-effective educational interventions based on a review of over 150 impact evaluations in education (Angrist et al. 2020). Enhancing implementation may be a particularly efficient use of resources for governments and for educational approaches that are designed for delivery at scale.

Table 3: Results of a Randomized Increase in Targeted Instruction

	Learning Outcomes			
	(1)	(2)	(3)	(4)
Treatment: Subgrouping by Similar Sub-Levels	0.190	0.225	0.211	0.208
	(0.098)	(0.099)	(0.087)	(0.084)
	[0.058]	[0.027]	[0.019]	[0.017]
Observations	1069	1069	1069	1069
Baseline Level Control	Yes	No	Yes	Yes
Baseline # Recognition Control	No	Yes	Yes	Yes
Region Fixed Effects	No	No	No	Yes

Notes: All standard errors are robust and clustered at the class level. P-values are reported in brackets. Learning gains are expressed in terms of standard deviations using the control group standard deviation.

7 Evidence Aggregation Accounting for Implementation

Our results thus far suggest that implementation plays an important role in both the mean and the variation in treatment effects across settings. We have also shown that implementation is not a fixed feature of a setting but can be exogenously changed. We now introduce a framework to more comprehensively account for implementation in evidence aggregation. Our model has two key elements. First, we consider uncertainty in implementation during the aggregation process. Just as there is uncertainty in treatment effects, there is uncertainty in implementation which must be modeled. Second, we consider multiple implementation variables together, including both take-up and fidelity. Our randomized trial in Botswana showed that implementation is not solely a fixed or selected property of an environment, but rather the degree of implementation can be exogenously changed. Thus, a quantity of interest is what treatment effects might be attainable at different levels of implementation.

We build on the Local Average Treatment Effects (LATE) framework (Imbens and Angrist 1994), which decomposes an observed ITT effect into the LATE or TOT multiplied by take-up. However, we now consider "compliance" to be a multifaceted implementation measure rather than a single binary indicator per person, and we consider partial vs full implementation on *the same population* (i.e. we do not have a notion of subpopulations of compliers versus not, but rather a continuum or fractional compliance score defined at the study level). We then bring this set of concepts to the evidence aggregation problem. We confront the issue that our expanded notion of the "degree of implementation" can include multiple measures which are potentially correlated with each other. For example, consider our two dimensions of interest: take-up and fidelity. Fidelity cannot occur without take-up, but fidelity still captures a notion of implementation that is further along the degree of implementation. Bayesian inference lends itself particularly well to handle this statistical challenge. Bayesian models also provide the usual benefit of being able to better handle the small sample challenges common to evidence aggregation settings, since priors can regularize estimates and variance parameters which often need to be estimated from a few studies or settings.

Thus, we develop an evidence aggregation model to account for the fact that implementation is both multifaceted and measured with uncertainty. We begin with the familiar setting of calculating, reporting, and aggregating TOT effects. Our model then introduces a multifaceted notion of implementation and its joint uncertainty within and across sites. Our model builds on existing frameworks accounting for implementation to achieve internal validity and extends these frameworks to account for implementation to inform external validity.

7.1 Building the joint aggregation model

Implementation can often be more complex than a proportion of individuals who received a fixed treatment. In the case of targeted instruction, students are typically expected to attend several lessons – thus, even the notion of "take-up" or attendance for one individual is not a binary but a fraction of the assigned lessons. Moreover, we would not consider the program to have been delivered if the students were not leveled or assessed correctly, or did not actually receive targeted instruction, even if they were physically present in the class. In such cases, modeling

compliance as a simple binary variable and conceiving of some individuals as "compliers" versus "non-compliers", may not capture the variation of interest. To reflect this in our data more accurately, we now conceive of treatment delivery or compliance as a proportion or continuum to which a program is effectively received as intended in a given setting as a whole. We denote this broader metric of the extent of implementation as $m \in [0, 1]$, and we will now discuss its measurement and construction as well as its use in our aggregation model.

To start, consider a fundamental insight of the LATE model: that an estimate of an observed effect combined with an estimate of the degree of implementation or compliance can be used to infer something about a latent or unobserved effect. In the canonical LATE framework, the observed ITT is decomposed into the proportion of compliers π_c and the Local Average Treatment Effect (LATE): $ITT = \pi_c LATE$. We will draw on this multiplicative structure, but now consider a case with varying degrees of implementation m for the same population, with this compliance or implementation factor defined at the level of the group. In our model, we call this potential effect that might be accessed in settings with full implementation the "Latent Treatment Effect (LTE)", denoted by θ and achieved at $m = 1$. If $m < 1$, there is less than complete or full implementation, which drives a wedge between what might have happened and what did happen. We call the effect we actually see at a given value of m as the "Realized Treatment Effect (RTE)". Thus, our implementation model uses this structure:

$$RTE = m\theta.$$

This equation expresses the idea that there is a single latent effect we can access fractionally. If $m = 1$ then the $RTE = LTE$, just as if the the full population were compliers in the LATE setting then the ITT = LATE. We assume effects increase monotonically with m.[19]

We apply this notion to evidence aggregation across settings. Consider a set of contexts $j = 1, 2, 3...J$. In each setting, there is a latent treatment effect of the program achievable if it is fully implemented denoted $\theta_j \in \mathbb{R}$. The extent of implementation in a setting j is denoted m_j, which is 0 when no component of the program is delivered to recipients, and $m = 1$ when the program is delivered as intended to all recipients. When implementation is only partially achieved, we expect to only receive a corresponding fraction of the total potential impact in that setting, so the Realized Treatment Effect RTE_j is the observed impact of the program in setting j, defined as the product $m_j\theta_j$. This model as a whole is defined by three sets of objects:

$$\{m_j\}_{j=1}^{J} \text{ where } m_j \in [0, 1] \; \forall j$$
$$\{\theta_j\}_{j=1}^{J} \text{ where } \theta_j \in \mathbb{R} \; \forall j$$
$$\{RTE_j \equiv m_j\theta_j\}_{j=1}^{J} \; \forall j$$

To aggregate evidence across the J settings, we build a hierarchical structure on top of this, adapted from the classical Rubin (1981) model. We assume that we observe the estimated

[19] It is conceivable that there are some cases where a 95% implemented program is actually less effective than a 70% implemented program, especially if one only considers a single dimension of implementation, and another dimension might offset and make the program better. However, by building a more general framework to consider multiple essential dimensions of implementation together, this assumption is likely to hold.

realized effect $\widehat{RTE}_j$ with some noise $\hat{se}_j$. We also observe an estimate of the implementation level, $\hat{m}_j$ with standard error $\hat{se}_{m_j}$. Then, we can use the following hierarchical likelihood structure:

$$\begin{aligned} \widehat{RTE}_j &\sim N(m_j\theta_j, \hat{se}_j^2) \\ \hat{m}_j &\sim N(m_j, \hat{se}_{m_j}^2) \\ \theta_j &\sim N(\theta, \sigma_\theta^2) \end{aligned} \tag{7.1}$$

To connect this model back to our earlier intuition, consider the definition and measurement of the implementation of targeted educational instruction programs. A researcher could define the implementation level purely as student take-up; in this case the recorded attendance rate of the classes would form the estimate $\hat{m}_j$. Given that our goal has been to broaden the concept of implementation beyond takeup, an important question is how to aggregate evidence when we have data on multiple aspects of program execution. The answer is to apply the above logic recursively: for example, attendance of students and instructors in program j is captured by a variable $m1_j \in [0,1]$ and fidelity of instruction is captured by another variable $m2_j \in [0,1]$. We assume that if only half the students and instructors show up, this dilutes the effect that the program can have in half, and if only half the instructors actually deliver targeted instruction, this dilutes the program effect in half again. To perform joint inference on all these factors, we use the following model:

$$\begin{aligned} \widehat{RTE}_j &\sim N(m1_j m2_j \theta_j, \hat{se}_j^2) \\ \widehat{m1}_j &\sim N(m1_j, \hat{se}_{m1_j}^2) \\ \widehat{m2}_j &\sim N(m2_j, \hat{se}_{m2_j}^2) \\ \theta_j &\sim N(\theta, \sigma_\theta^2). \end{aligned} \tag{7.2}$$

This "2-factor" form of the model allows us to understand both whether implementation matters and which aspects of implementation matter. Conceptually, the model above can be recursively expanded even further.

7.2 Estimation and Inference

We consider estimation and inference in a setting where the studies are randomized trials, offering unbiased estimation of treatment effects. We assume that reported treatment effects are estimated with Gaussian sampling error centered around the true effect, and that corresponding standard errors are reported. This is the typical assumption underlying the computation of standard Gaussian confidence intervals and p-values in the applied economics literature. We assume there is access to implementation data $\{\hat{m}_j\}_{j=1}^J$. We allow for this to be defined by the original implementors and study authors. What matters most for our purposes is that the researchers are able to define, estimate and report m_j along with accompanying standard errors.

We now consider estimation and inference, particularly on the hyperparameters: the hypermean θ, which is the average latent treatment effect in the population of effects from which the studies come, and the hypervariance σ_θ^2, which captures the degree of heterogeneity in these effects across settings. Because we typically aggregate evidence from few studies, particularly

here given our inclusion criteria restricts us to only aggregate treatment effects from RCTs, we will use Bayesian inference. To do so we will need priors on the hyperparameters. We use the following weakly informative priors:

$$\theta \sim N(0, 10) \tag{7.3}$$

$$\sigma_\theta^2 \sim halfNormal(0, 10). \tag{7.4}$$

We will fit the model using the Hamilton Monte Carlo algorithm in Stan. Given that we intend to apply the model to small J, sometimes with the necessity for multiple deconvolutions of the implementation factors, we conduct a basic check that the inference procedure is robust for a reasonable number of iterations. We assess baseline model performance via simulation. Appendix Table A4 and Table A5 show that the model passes this basic check.[20]

7.3 Implementation evidence aggregation model results

We investigate effects considering both take-up (measured by the presence of students, teachers, and teaching materials) and fidelity to program quality (measured by students being grouped by learning level in the classroom).

We first estimate Latent Treatment Effects (LTE) by fitting the single-factor implementation model (equation 7.1) to our data. Table 4 shows the results for all studies in panel A, teacher-delivery method studies in panel B, and volunteer-delivery in panel C. We show the posterior mean along with 5 posterior quantiles to give the full sense of the distribution, and report the Rhat criterion as a convergence diagnostic. As the results show, the latent treatment effects for all studies are both much larger than the average realized effects and more generalizable, but the difference is much more marked for volunteer studies. The average latent treatment effect for volunteer-delivered programs is around 0.5 standard deviations, compared to around 0.2 SDs for teacher-delivered programs. This implies that latent treatment effects for teachers are likely to be positive in most settings, whereas for volunteers they are always large and positive.

These latent treatments effects are somewhat smaller than the direct TOT analysis results from previous sections. In the case of volunteers, there is a 0.1 SD reduction in the estimated average latent effect. This is not due to priors, as our Bayesian analysis of TOTs in the previous section largely replicated the frequentist results. Instead, this reduction in the estimated effect seems to be due to the uncertainty on implementation. Our initial analysis, which first computed the TOTs and then aggregated, conditions on the estimated uncertainty when aggregating, rather than incorporating the full data and propagating the full uncertainty.[21] Our model places bounds on the implementation factors' values without imposing a linear probability model, which

[20] One drawback of our current approach is that we do not explicitly consider correlations between implementation levels m_j and latent effects θ_j – this amounts to assuming that m_j carries no additional information about θ_j after they have been deconvolved, such that places with higher m_j are not systematically different in terms of their θ_j. This assumption simplifies the models enough to make them tractable on small data sets. Accounting for the correlation is still possible ex-post in the joint posterior, it just is not used to improve inference on any other parameters. With larger data sets, a richer model with a joint hierarchical structure placed on (m_j, θ_j) could be suitable.

[21] We also note that the level of takeup in our data ranged from 8 percent to 90 percent; given this substantial range, the linear probability model that underpins the Wald Estimation of the TOT effects is likely to be stressed by the data at least in some cases, and has perhaps been unduly influenced by extreme results.

Table 4: Model with Takeup as Implementation factor: Posterior Distribution on Effects

	mean	2.5%	25%	50%	75%	97.5%	Rhat
Panel A: Latent Treatment Effects (All)							
Hypermean	0.425	0.244	0.362	0.418	0.478	0.639	1.003
HyperSD	0.208	0.070	0.134	0.186	0.257	0.478	1.000
Panel B: Latent Treatment Effects (Teacher)							
Hypermean	0.225	-0.144	0.155	0.221	0.298	0.575	1.014
HyperSD	0.217	0.005	0.054	0.133	0.288	0.935	1.004
Panel C: Latent Treatment Effects (Volunteer)							
Hypermean	0.512	0.298	0.444	0.499	0.572	0.766	1.004
HyperSD	0.187	0.018	0.087	0.154	0.252	0.552	1.003

Note: This inference is generated by running 4 chains of HMC for 2000 iterations. Rhat is a diagnostic criterion for MCMC convergence with multiple chains in which a value close to 1 indicates good mixing.

may be another reason why we see more uncertainty in these results. Given the small sample of studies, the more tempered conclusion seems likely to be more accurate, as underpowered inference tends to produce overestimation (Gelman and Carlin, 2014).

We now consider the data on program fidelity as another important aspect of the implementation of targeted instruction. Although we only have this data for 3 study-arms and we view the results below as suggestive, the implementation model still performed well in simulations at $J = 3$ (see table A5). Table 5 shows results from the 2-factor model with fidelity as the implementation level in Panel A, and Panel B shows the results considering takeup and fidelity jointly. In both cases, we see large latent treatment effects. The results in Panel A show the average latent treatment effect is now around 0.85 SDs, with a hyperSD around 0.38. The inference in Panel B aligns with Panel A and shows even larger latent treatment effects once fidelity is accounted for. This indicates an additional 0.4 SD improvement in the latent treatment effects over considering take-up alone, revealing the importance of considering multiple aspects of implementation.

Table 5: Implementation Model including Fidelity and Takeup: Posterior Effect Distributions

	mean	2.5%	25%	50%	75%	97.5%	Rhat
Panel A: Fidelity on TOT							
Hypermean	0.850	0.348	0.729	0.836	0.938	1.503	1.016
HyperSD	0.384	0.008	0.092	0.222	0.475	1.836	1.011
Panel B: Fidelity and Takeup Jointly							
Hypermean	1.169	0.263	0.991	1.132	1.316	2.226	1.004
HyperSD	0.657	0.011	0.120	0.338	0.851	3.068	1.003

Note: This inference is generated by running 4 chains of HMC for 8000 iterations. Rhat is a diagnostic criterion for MCMC convergence with multiple chains in which a value close to 1 indicates good mixing.

8 Conclusion

Our analysis demonstrates the importance of quantifying program implementation with as much care as we typically quantify program effects. In our data, we find that implementation factors explain most of the variation in the effects of targeted instruction programs across settings. Here, external validity can be enhanced by simply reporting treatment-on-the-treated effects or LATE estimates in addition to the more typically reported intention-to-treat effects. Further insight can be gained by using an evidence aggregation model that formally accounts for uncertainty in implementation and incorporates different aspects and measures of implementation.

Our results also demonstrate that implementation does not have to be taken as given by NGOs, nor by researchers and policy analysts. It is not a fixed feature of an environment; it can be changed in practice, as our RCT in Botswana shows. Similar analyses could readily be conducted across additional programs in the education sector beyond targeted instruction and might also prove relevant in additional sectors beyond education. A recent study finds similar results in the context of cancer screening trials (Angrist and Hull 2023), though the converse is found in other settings (de Chaisematin and Deeb, 2024). Further study of program implementation seems promising, motivating the collection of data on takeup and fidelity at a much more extensive level than is currently practiced.

Our results suggest several avenues for future research on targeted educational instruction in particular. Most importantly, it appears targeted instruction has large and generalizable effects across low- and middle-income settings, suggesting the approach could make a substantial dent in the learning crisis as it scales, as long as implementation take-up and fidelity are high and accounted for. In terms of future research, our results suggest that frontier research should focus not on entirely new approaches in education, but rather on optimizing implementation of known approaches at scale. Implementation should be conceived of broadly, including delivery models, such as teacher or volunteer instruction, and degrees of implementation, such as takeup and fidelity. Developing new theory and practice grounded in richer notions of program implementation may be an important avenue for future work.

An open question is why targeted instruction approaches have been so consistently effective in low- and middle-income (LMIC) contexts yet related approaches more mixed evidence in high-income settings. This could be relevant to the literature on "differentiated instruction" (Tomlinson 2014), a term used to describe approaches similar to targeted instruction in high-income settings and which have found mixed effects. Program implementation seems likely to play a role: a recent systematic review highlighted that in many high-income settings "differentiated instruction has been operationalized in many different ways" (Smale-Jacobse et al. 2019). For example, targeted instruction approaches tested in LMICs typically involve dynamic regrouping of students every few weeks, ensuring instruction is always targeted to children's learning levels. In contrast, in many high-income countries, differentiation involves once-off or more sporadic learning assessment and grouping of students, likely leading to instruction which is not as well targeted to children's learning level. Quantifying the degree of fidelity in targeting approaches in future experimental studies might shed light on this and bridge the gap between results across settings.

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A Appendix Tables

Table A1: Representativeness - Comparing those who implement to a broader sample

	Compliers (relative to broader sample)				
	(1)	(2)	(3)	(4)	(5)
	UP 10 Day Camps	UP 20 Day Camps	Haryana	Bihar School	Bihar Summer Camp
Female	-0.006	-0.004	0.017	-0.017	-0.037
	(0.011)	(0.011)	(0.012)	(0.022)	(0.020)
	[0.574]	[0.743]	[0.164]	[0.444]	[0.071]
Age (years)	0.003	0.002	-0.006	-0.002	-0.006
	(0.009)	(0.010)	(0.007)	(0.009)	(0.005)
	[0.782]	[0.795]	[0.425]	[0.777]	[0.247]
Grade at baseline	-0.001	-0.008	0.009	-0.014	0.015
	(0.011)	(0.012)	(0.009)	(0.020)	(0.011)
	[0.940]	[0.502]	[0.324]	[0.471]	[0.185]
Score on base read ASER test	0.004	0.003	-0.002	0.020	0.004
	(0.008)	(0.009)	(0.005)	(0.018)	(0.010)
	[0.659]	[0.743]	[0.667]	[0.261]	[0.716]
Score on base math ASER test	-0.011	-0.012	0.004	0.019	0.002
	(0.008)	(0.009)	(0.008)	(0.020)	(0.011)
	[0.181]	[0.205]	[0.589]	[0.356]	[0.887]

Notes: All standard errors are robust and clustered at the class level. P-values are reported in brackets. This table compares the subset who implement in a given geography and treatment to the rest of the sample along a host of baseline covariates such as gender, age, grade, and baseline level of leaning. We run a regression with a dummy for those who implement as the dependent variable and a vector of covariates as the independent variables. This enables us to assess representativeness the subset of the sample who are compliers. We include a subset of studies with uniform baseline variables for comparability.

Table A2: First stage F-statistics

	First stage take-up rate F-statistics				
	(1)	(2)	(3)	(4)	(5)
	UP 10 Day Camps	UP 20 Day Camps	Haryana	Bihar School	Bihar Summer Camp
Treatment	456.39	237.56	189.49	128.49	46.94

Notes: This table includes F-statistics on take-ups rates for the first-stage used for instrumental variable 2SLS calculation for the treatment on the treated estimates. We include a subset of studies with uniform baseline variables for comparability.

Table A3: Randomization Balance Tests

	Baseline Learning Outcomes			
	(1)	(2)	(3)	(4)
	Level	Level	# Recognition	# Recognition
Treatment: Subgrouping by Similar Sub-Levels	0.052	0.094	-0.029	-0.024
	(0.329)	(0.340)	(0.136)	(0.142)
	[0.874]	[0.782]	[0.831]	[0.869]
Observations	1080	1080	1080	1080
Region Fixed Effects	No	Yes	No	Yes

Notes: This table presents balance tests for the randomized trial conducted in Botswana to test the degree of implementation by enhancing fidelity to targeted instruction. All standard errors are robust and clustered at the class level. P-values are reported in brackets. Baseline learning is expressed in terms of standard deviations using the control group standard deviation.

B Appendix Figures

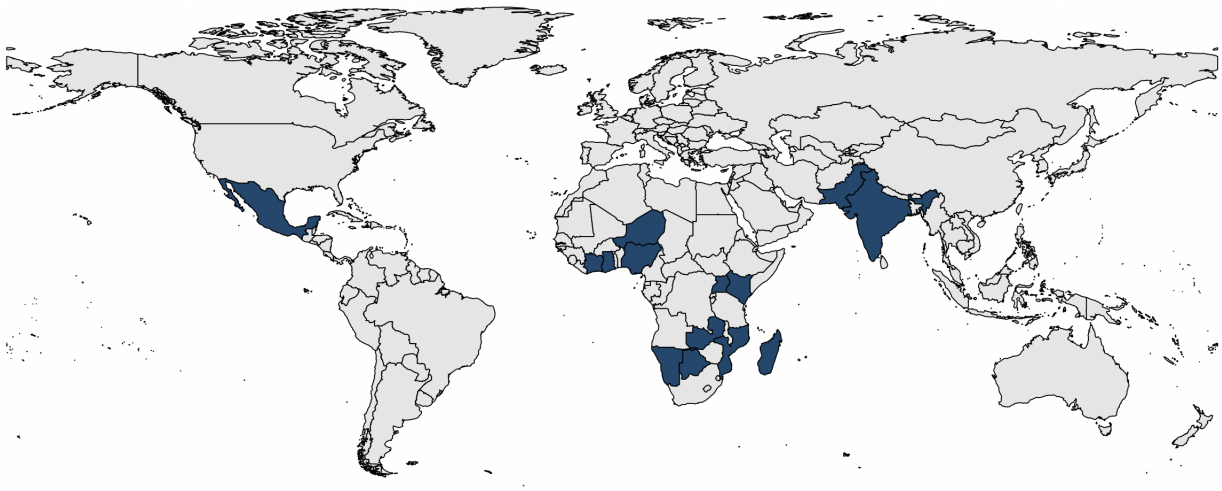


Figure A1: Active targeted instruction scale-up efforts are ongoing in Botswana, Cote D'Ivoire, Ghana, India, Kenya, Madagascar, Mexico, Mozambique, Niger, Nigeria, Pakistan, Uganda, and Zambia. For more information see https://www.teachingattherightlevel.org/)

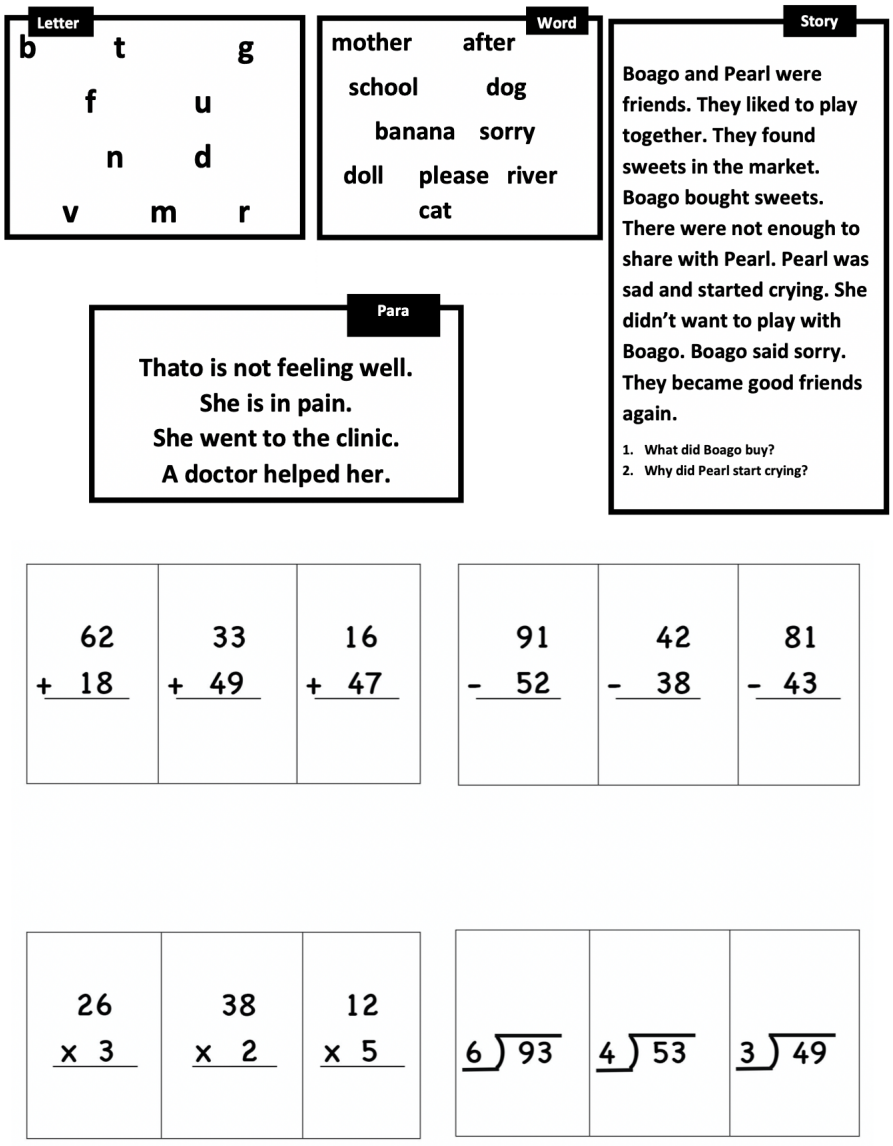


Figure A2: ASER assessment examples

C Additional details on Bayesian evidence aggregation

C.0.1 The Hierarchical Modeling Approach

We start with the canonical Rubin (1981) "Eight Schools" Bayesian hierarchical model. This model has been extensively used in the literature and considers a set of J total estimated treatment effects $\hat{\theta}_j$ and their standard errors $\hat{se}_j$ (Rubin 1981, Gelman et al 2004, Meager 2019). The estimates are typically assumed by authors of empirical research papers to be Normally distributed around the true effects θ_j, since they use a consistent estimator and invoke the Central Limit Theorem. These assumptions underlie the computation of confidence intervals and p-values in frequentist research papers, imposing no additional structure on the original papers' analyses. The hierarchical component of the model additionally posits that these effects are Normally distributed around some true average or "hypermean" effect θ, with some "hyper standard deviation" or "hyperSD" σ_θ which governs their dispersion around the true effect. The resulting hierarchical likelihood is written as follows:

$$\begin{aligned} \hat{\theta}_j &\sim N(\theta_j, \hat{se}_j) \\ \theta_j &\sim N(\theta, \sigma_\theta^2) \end{aligned} \tag{C.1}$$

This parametric model is more general than it appears: for example, if the hyperSD is set to 0 this model nests classical Frequentist fixed-effects meta-analysis. The model can be estimated in a frequentist manner, although it can be easier for this model to be estimated using Bayesian methods. The model's performance has been extensively discussed in Gelman et al. (2004) and has good frequentist properties, including attaining nominal coverage rates for the posterior credible intervals; i.e. the central 95% posterior intervals contain the true parameter 95 percent of the time. The key assumption embedded in the model is that of exchangeability between the effects being studied, which is a weaker form of the classical i.i.d assumption, and plausible in meta-analytic settings (Meager 2019).

To estimate this likelihood model in a fully Bayesian manner it is necessary to include a prior – that is, adding a prior distribution to the hyperparameters. Following Gelman et al. (2004) and Meager (2019), we use weakly informative priors as a default approach: this imposes some structure without unduly influencing the posterior results. For the hypermean, we tend to center our prior at zero with a wide uncertainty interval, reflecting the principle that researchers ought to have as their "null hypothesis" the contention that an untested intervention or policy should be considered most likely to have no impact until proven otherwise by the data. For the hypervariance, we use half-Normal or half-Cauchy priors as suggested in Gelman and Hill (2007), which allows for large variation in effects across settings. Priors can improve overall estimation by making a favorable bias-variance tradeoff even when the prior information is incorrect. It is also possible to encode more substantive information in priors, such as basing priors on economic theory or contextual knowledge. We conduct analysis with informative priors in Appendix 6 and 7 and in Section H.1. This provides both results of substantive interest as well as robustness checks on the strength of patterns in the data in the face of strong priors.

C.0.2 Meta-regression within the Hierarchical Framework

We use meta-regression to explore the role of program-level covariates. Meta-regression is straightforward within the Bayesian hierarchical approach: in the Rubin (1981) model one need only replace the hypermean θ with a conditional hypermean expression in the style of linear regression. This can be implemented using the following model. Given a set of K contextual factors and covariates, defined by vector X_j for site j, one can specify a parameter β such that the expected value of the effects θ_j is the conventional regression surface, and hence $E[\theta_j] = X_j\beta$. This is implemented via the hierarchical meta-regression likelihood below:

$$\begin{aligned} \hat{\theta}_j &\sim N(\theta_j, \hat{se}_j) \\ \theta_j &\sim N(X_j\beta, \sigma_\theta^2) \end{aligned} \tag{C.2}$$

While the model is fully parametric, it can be generalized: If one were to discard information about sampling variation and assume $theta_j = \hat{\theta}_j$ then the model above corresponds in expectation to classical Frequentist meta-regression. This is because the kernel of the Gaussian likelihood corresponds to the Ordinary Least Squares objective function.

One important note about this model is that in the context of few studies, say J < 15, and many covariates of interest, say K > 3, then the estimation of the regression coefficients β becomes very challenging and noisy. Moreover, the problem may be masked due to the risk of overfitting. However this issue can be addressed via the use of a Machine Learning technique known as regularization (Hastie et al 2009). Regularization involves the incorporation of a penalty function to prevent an estimation procedure from freely wandering around the parameter space. Classic examples include Ridge Regression, which imposes a squared penalty on the size of the estimated regression coefficients, and Lasso, which imposes an absolute value penalty on the same quantity. Within the Bayesian context, it is natural to use the priors to impose the penalty.[22] We use this penalty throughout our hierarchical meta-regression.

C.0.3 Assessing Heterogeneity and Generalizability

There are several approaches within the Bayesian framework to assessing heterogeneity in effects and generalizability of results. The hyperSD parameters capture the population variation in effects and deserve particular attention. We report the results on the hyperSD throughout, which is usually treated as the fundamental parameter in Bayesian hierarchical models (Gelman et al. 2004). However, it is challenging to know how large or small a particular hypervariance estimate is, or how best to interpret it. Thus, we provide two additional metrics of heterogeneity: the frequentist I-squared metric, and the Bayesian posterior predictive distribution.

The I-squared metric measures the percentage of the total variation in estimated effects around the hypermean that is due to genuine variation in true effects, rather than to sampling variation that causes estimates to vary more than the true effects. This is the reciprocal metric to the conventional Bayesian pooling factor discussed in Meager (2019) and Gelman and Pardoe (2006), which measures the percentage of total variation in effects attributable to within-study sampling variation. When I-squared is high, this indicates true treatment effect variation is

[22] As discussed in Hastie et al (2009), in Bayesian analysis a Gaussian prior on the regression coefficients centered at zero is analytically identical to a Frequentist Ridge Regression penalty.

higher than sampling variation – that is, the heterogeneity across settings dominates the uncertainty within settings. This makes extrapolation across settings challenging and suggests low generalizability. In this case, pooling of effects across studies is low. Conversely, when I-squared is low, sampling variation is larger than true treatment effect heterogeneity across settings. This corresponds to relatively high external validity, and the Bayesian hierarchical model will have higher pooling factors and perform more "partial pooling" of effects.

Posterior predictive distributions provide another metric to assess heterogeneity in effects across settings. These distributions capture the uncertainty about the hypothetical treatment effect in the next study. In the Rubin (1981) model, when defining the posterior distribution of these hyperparameters as $F(\theta, \sigma_\theta)$, the posterior predictive distribution for the next effects is:

$$\theta_{J+1} \sim N(\theta, \sigma_\theta^2 | F(\theta, \sigma_\theta^2). \tag{C.3}$$

If one uses an aggregation approach that does not explicitly measure heterogeneity in effects across programs or studies, such as a fixed-effects meta-analytic model, the posterior predictive distribution is simply the posterior distribution of the hypermean itself. This is because there is no specified heterogeneity in effects across studies in such a model and thus no quantification of the cross-study extrapolation error. Hence, the extent to which the Bayesian hierarchical posterior predictive distribution is wider than the posterior distribution on the hypermean indicates the extent of heterogeneity in the effects. Posterior predictive distributions capture how heterogeneity across settings impedes or enables our ability to extrapolate evidence to the future targeted instruction intervention. This is natural metric of generalizability, linking the present evidence base to future settings and is presented in Figure A4.

D Bayesian Meta-regression Results

To systematically analyze which factors mediate treatment effects of targeted instruction, we turn to meta-regression. We consider various factors, with a focus on baseline learning levels and implementation delivery model, two of the most probable mediators of large effects. In Figure A3, we show the results of fitting these models for both the ITT and TOT effects alongside the original results of the basic aggregation, as well as the results of meta-regression models. We present the inference for each study as well as the pooled estimate (the bottom row of the graphic), which is the average effect of targeted instruction across all settings.

As Figure A3 shows, for the ITT, running meta-regression models conditioning on either or both covariates of interest has little impact on the inference. The basic model's findings are confirmed by the more advanced models: the ITT effects are positive yet substantially heterogeneous across settings. By contrast, the TOT effects are now less heterogeneous, and what heterogeneity is present is substantially explained by these covariates. The TOT model conditioning on implementation delivery model has both the largest average effect of 0.53 standard deviation, as well as the most precise inference (the pink bar on the bottom line of the figure). Examining each study in turn, we can see visual evidence that the teacher camps, Balshaki camps and first UP camps have their estimated TOT effects somewhat revised upwards. We check robustness and confirm that the results are not contingent on any single study in the Appendix using a leave-one-out robustness check in Section H.2.

Conditioning on baseline educational performance surprisingly does little to improve precision. One might expect either a negative correlation due to ceiling effects (high-performing students or schools already perform well, so benefit less from remedial classes) or a positive correlation due to selection (students in high-performing schools know how to learn, so benefit more from remedial classes). Yet we see little evidence of any substantial correlation in the data, with this covariate exerting little influence. It is possible that in this data set, all students are so far behind the curriculum that all benefit from targeted instruction, and any additional variation has only a marginal effect. It is also possible that implementation delivery model and the degree of implementation simply play such a large role such that other factors are minor in comparison.

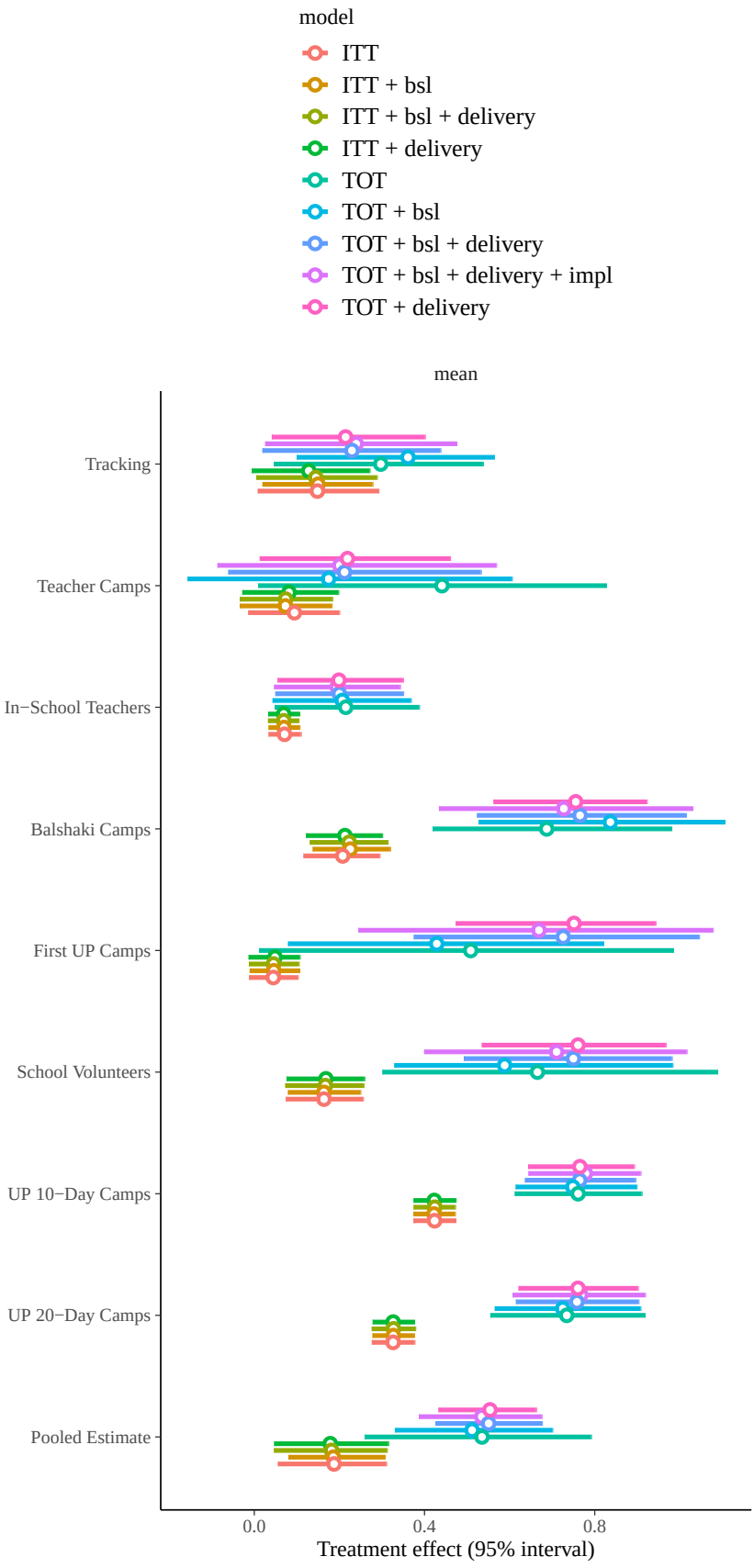


Figure A3: Summary of Bayesian aggregation results across multiple models

E Posterior Predicted Treatment Effects

To understand how our statistical aggregation results translate into extrapolation to future policy settings, we now examine the posterior predicted distributions of the next comparable study's ITT and TOT effects respectively. Figure A4 shows the uncertainty interval of the predicted effect of targeted instruction in the next setting, labelled "predicted draws", and for comparison also shows the uncertainty interval on the average effect of targeted instruction across settings.

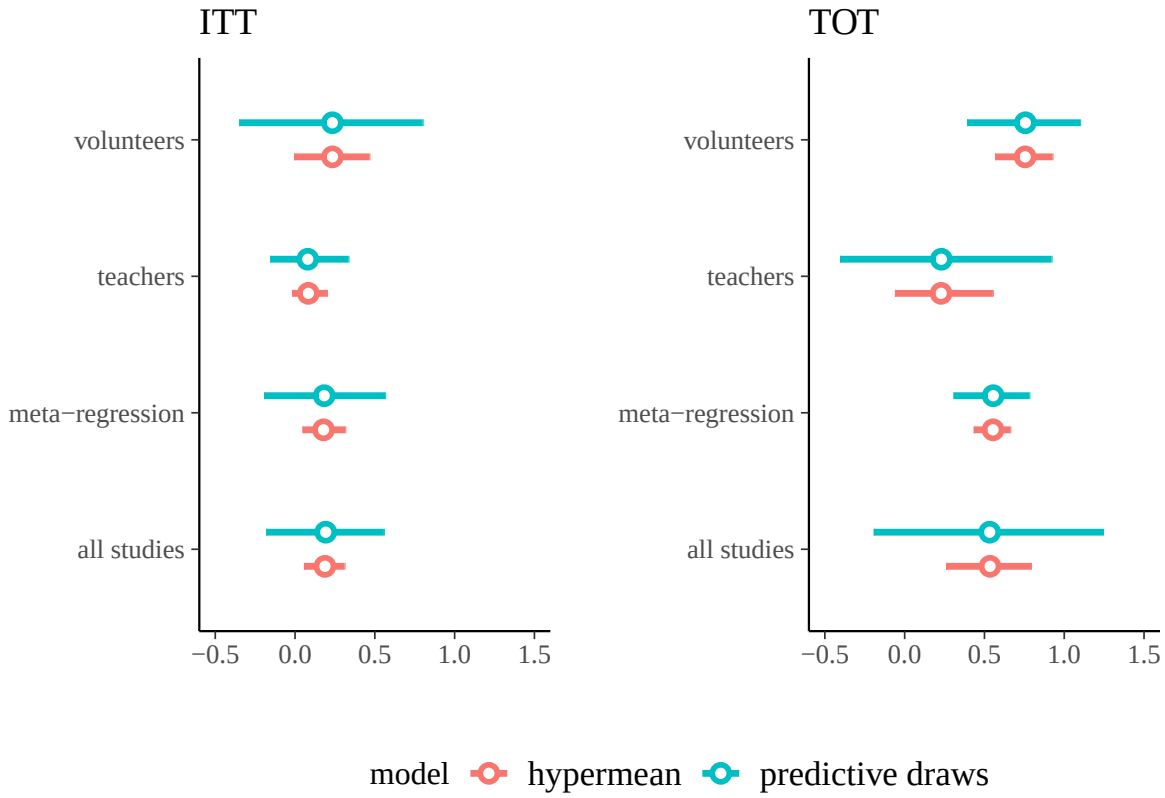


Figure A4: Posterior Predictive Distributions of Future Effects

To interpret figure A4, recall that if the effect of targeted instruction were homogeneous in all settings, the red and green distributions would be the same because the average effect would then be the predicted effect everywhere. Classical fixed-effects meta-analysis does not distinguish between these two quantities; in that context, the posterior uncertainty on the hypermean is the posterior uncertainty on the predicted effect. But in the presence of heterogeneity of effects across settings, there is a fundamental extrapolation error when attempting to use the mean to predict the specific effect in any setting, which ought to be reflected in greater prediction uncertainty; this can be captured in the hierarchical model. Our results show that there is heterogeneity in both the raw ITT and TOT effects of targeted instruction, but the gap is much smaller for the meta-regression model on the TOT effects, confirming that accounting for contextual factors eliminates much of the heterogeneity across settings. The figure further shows that accounting for differential delivery mechanism seems to capture some of the variation in effects in the TOT, as the uncertainty is lower on the split models than on the average of all studies, even though the average is estimated from more data.

We observe a few patterns. First, average effects and effects for teachers do not consistently

have positive effects in all posterior distributions; only volunteer TOT estimates do. This is likely due to volunteer TOT effects being both larger on average and, crucially, more homogeneous across studies. In short, our analysis finds strong evidence that volunteer-lead targeted instruction interventions have a generalizably large and positive impact. By contrast, though the average TOT and ITTs effect for all targeted instruction programs is positive in each subcase and in each model, there is too much heterogeneity across study contexts to rule out the potential for negative effects in a model that allows for effect distributions to be symmetric (as all classical meta-analytic models do).

Figure A4 also shows that for TOT effects, meta-regression substantially improves the precision of the inference on the hypermean and the posterior predictive draws. As the right panel shows, the posterior predicted TOT effect from the meta-regression model is even smaller than the posterior hypermean of the basic Rubin (1981) model – that is, these covariates more than compensate for the original extrapolation error that one would have attained in the basic Bayesian or indeed Frequentist aggregation exercise. Moreover, the variation within the delivery groups (teachers and volunteers) is larger than the variation remaining once one conditions on this type of study (meta-regression). These results formally confirm our earlier finding that implementation (TOT vs ITT) and delivery mechanism (volunteer vs teachers) substantially explains variation in TOT results, and has a key role to play in predicting the relative success of targeted instruction interventions across settings.

E.1 Full Posterior Predicted Distribution Graphics

Figure A5 shows the basic model in red. Meta-regression models conditioning on baseline are shown in green, on implementation delivery model in purple, and on both in blue. As the figure shows, the predicted effect is virtually identical in each case. We examine the same graph for the TOT, and here find that the posterior distribution of predicted effect in the next setting is substantially more precise for the model conditioning only on delivery (shown in purple). This result confirms that the remaining heterogeneity in effects is largely predicted by two implementation factors: implementation degree (e.g. TOT) and implementation delivery model.

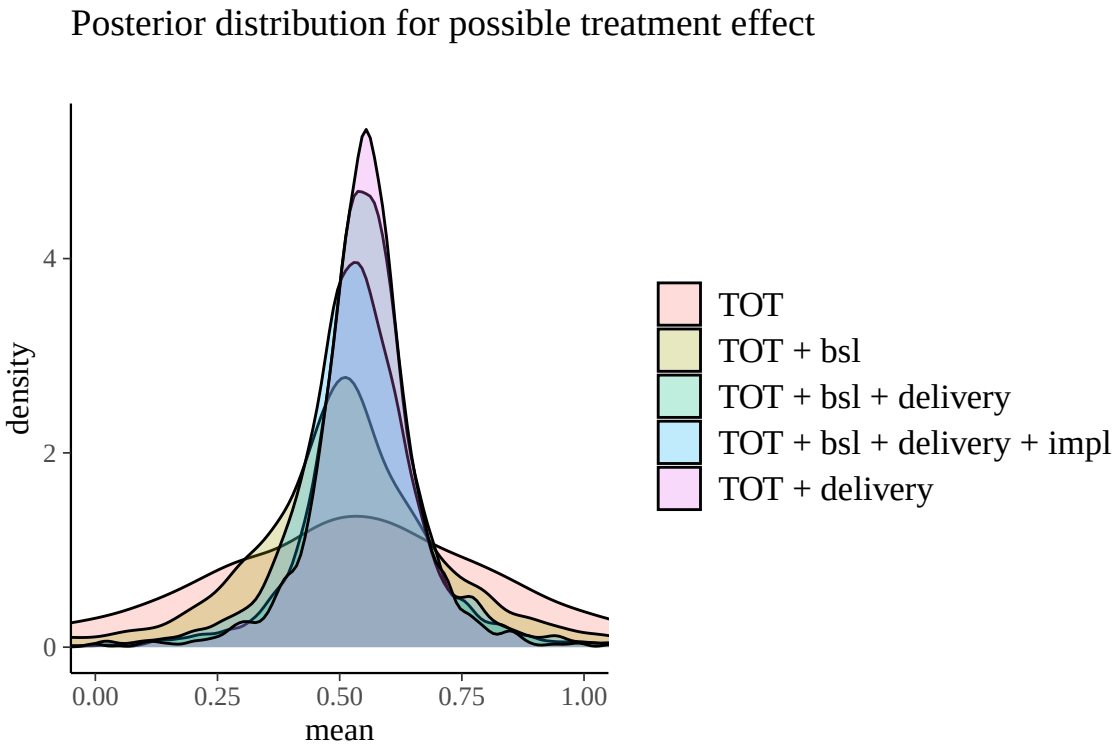


Figure A5: Full posteriors predictive distributions under different models

F Bayesian Implementation Model: Estimation Performance

We now show via simulation that reliable estimation and inference is possible using both the one factor model and the two-dimensional implementation model even when J is very small.

We consider data sets of size $J = \{3, 5, 8, 15\}$, and for each case we run 1000 simulations from the model above, where the true hypermean is drawn from the prior $N(0, 10)$ and the true hyperSD is drawn from the prior $halfNormal(0, 5)$. We draw the J standard errors on the realized treatment effects from a uniform distribution from 5 to 20. We draw the J true implementation factors from a uniform distribution on [0.1, 0.9] which is roughly the range in our data set and we draw their standard errors from a uniform distribution on [0.002, 0.02] because this is roughly their magnitude in our data set (and to avoid crossing the parameter boundaries). In each case we record the root mean squared error of the posterior mean and posterior median of each of the hyperparameters (θ, σ_θ), as well as the true frequentist coverage of the 50% and 95% posterior credible intervals across all the simulations for each case.

Each model is fit using a Hamiltonian Monte Carlo algorithm implemented via Stan in R. We run 4 chains of 4000 iterations in each case.

The results for the single implementation factor model (equation 7.1) are shown below in Table A4, and in Table A5 for the 2-factor implementation model (equation 7.2). As the results show, the models achieve approximately nominal coverage with very low root mean squared error even for small J. Of course, in applications we do not have the luxury of knowing the generating process, but this provides assurance that good inference is at least theoretically possible even with very small data sets.

Table A4: 1-Factor Implementation Model Performance in Simulations

Studies	Parameter	RMSE (Mean)	RMSE (Median)	50% CI Coverage	95% CI Coverage
J = 3	Hypermean	6	6	0.501	0.958
	HyperSD	4	4	0.497	0.949
J = 5	Hypermean	6	6	0.502	0.956
	HyperSD	4	3	0.493	0.950
J = 8	Hypermean	5	5	0.546	0.971
	HyperSD	4	3	0.503	0.941
J = 15	Hypermean	4	4	0.512	0.968
	HyperSD	4	3	0.524	0.945

Table A5: 2-Factor Implementation Model Performance in Simulations

Studies	Parameter	RMSE (Mean)	RMSE (Median)	50% CI Coverage	95% CI Coverage
J = 3	Hypermean	7	7	0.499	0.942
	HyperSD	2	2	0.458	0.949
J = 5	Hypermean	7	7	0.503	0.945
	HyperSD	2	2	0.503	0.956
J = 8	Hypermean	6	6	0.510	0.957
	HyperSD	2	2	0.474	0.950
J = 15	Hypermean	5	5	0.514	0.949
	HyperSD	2	2	0.511	0.962

G Additional Bayesian Models and Results

G.1 Joint Aggregation Model

Since we have access to baseline information about each of the TaRL studies, we can go further than the basic Rubin (1981) model and employ a joint aggregation exercise that leverages this baseline information in order to improve precision and inference. We specify a joint hierarchy on the control group means and treatment effects in each TaRL study, as follows:

$$\begin{aligned} \hat{\mu}_k &\sim N(\mu_k, \sigma^2_{\mu_k}) \\ \hat{\tau}_k &\sim N(\tau_k, \sigma^2_{\tau_k}) \\ \begin{pmatrix} \mu_k \\ \tau_k \end{pmatrix} &\sim N\left(\begin{pmatrix} \mu \\ \tau \end{pmatrix}, V\right) \text{ where } V = \begin{bmatrix} \sigma^2_\mu & \sigma_{\tau\mu} \\ \sigma_{\tau\mu} & \sigma^2_\tau \end{bmatrix} \forall\ k. \end{aligned} \tag{G.1}$$

This joint model incorporates a correlation parameter between the baseline or control group mean and the treatment effects, which can improve precision and estimation overall if such a correlation is present. In other respects it is identical to the classical Rubin (1981) model. This model, developed by Meager (2019), is sometimes referred to as the "mu and tau" model, as in previous literature the effect of the program was labelled with the Greek letter τ rather than θ (see for example Gelman et al 2004). This model was shown to substantially improve precision and inference in the microcredit aggregation setting, and is thus worth incorporating into our main analysis in the hope of similar gains to estimation performance (see Meager 2019 for more details).

The results of this model are shown below in figure A6. They broadly confirm the Rubin (1981) results shown for comparison.

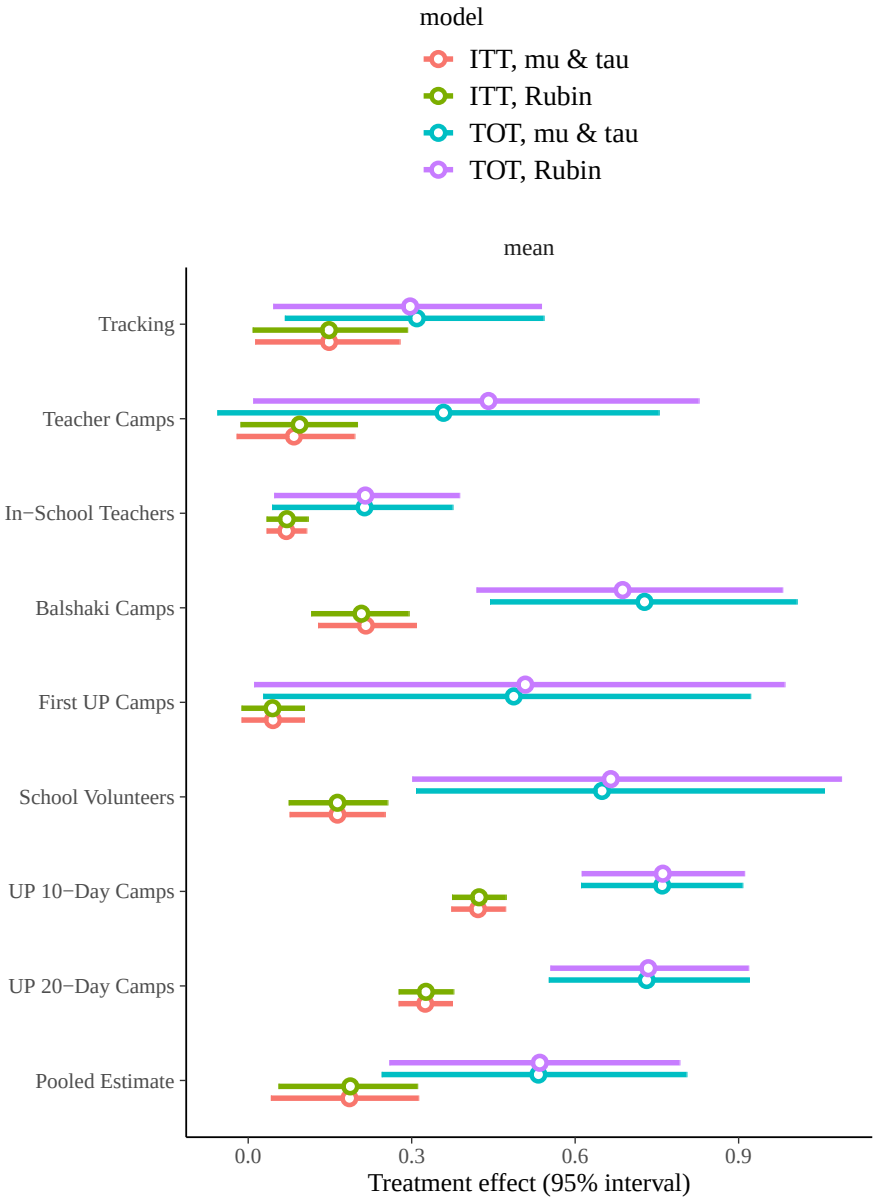


Figure A6: Rubin (1981) model vs joint "mu and tau" model

G.2 Bayesian Pooling Factors

We now display the pooling factor for each study in the simple Bayesian hierarchical model. Table A6 and table A7 show the mean estimated pooling factor along with the lower and upper bounds of the 95% posterior interval. The intervals are moderately tight and while both the Intention to Treat results and the Treatment on Treated results are heterogeneous across settings, the ITT is much more heterogeneous (i.e. with smaller pooling factors). The hierarchical models therefore perform much less partial pooling on the ITTs, even accounting for posterior uncertainty about these pooling factors themselves.

Table A6: ITT Effects: Estimated Pooling Factors

	2.5%	mean	97.5%
Balshaki Camps	0.020	0.092	0.202
First UP Camps	0.008	0.040	0.094
Tracking	0.053	0.207	0.405
Teacher Camps	0.032	0.135	0.285
School Volunteers	0.021	0.095	0.209
In-School Teachers	0.003	0.017	0.040
UP 10-Day Camps	0.006	0.028	0.067
UP 20-Day Camps	0.006	0.031	0.072

Table A7: TOT Effects: Estimated Pooling Factors

	2.5%	mean	97.5%
Balshaki Camps	0.054	0.251	0.544
First UP Camps	0.233	0.600	0.865
Tracking	0.036	0.186	0.441
Teacher Camps	0.131	0.445	0.759
School Volunteers	0.122	0.428	0.745
In-School Teachers	0.016	0.095	0.257
UP 10-Day Camps	0.013	0.078	0.217
UP 20-Day Camps	0.021	0.119	0.310

H Further Robustness

H.1 Prior Robustness results

Figure A7 shows that for both the hypermean and hyperSD (heterogeneity in effects across settings) even very strong priors cannot substantially influence the inference on the ITT results. The evidence on the positive impact of targeted instruction at the school level is extremely strong. This contrasts somewhat to the TOT results, which are somewhat more influenced by the priors – this is because, as discussed earlier, the TOT effects are estimated with greater uncertainty within each study and therefore less able to overcome pessimistic priors. However, the strength of the TOT evidence on a positive average (hypermean) is shown in the compensating pattern in the hyperSD; if the hypermean is forced down closer to zero, the hyperSD is forced upwards to compensate for evidence of large and positive effects in some studies. Moreover, even with the most pessimistic prior, the Bayesian models report more than 75% chance of a positive effect of targeted instruction. Overall, therefore, the evidence on the positive impact of targeted instruction is strong.

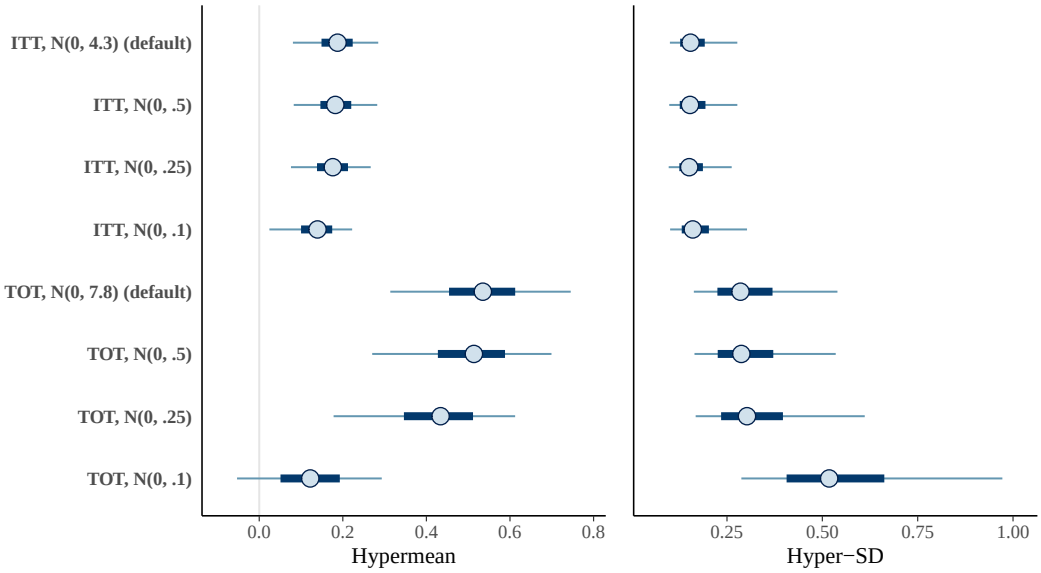


Figure A7: Additional Prior Robustness Checks

We examine the results of imposing a stronger theory-driven regularization of the correlation between baseline educational performance and the treatment effect of targeted instruction towards zero. The competing theoretical mechanisms suggest we should expect a small correlation; this corresponds to expecting or favoring independence or zero-off-diagonal terms in the variance-covariance matrix. This is implemented via the use of an LKJ Correlation prior distribution on the variance-covariance matrix V from the joint aggregation model described earlier in the Appendix. The LKJ distribution is a distribution over the space of correlation matrices, parameterized by a "concentration parameter" that can take any positive value (see Meager 2019 and Gelman and Hill 2007 for more information). If the concentration parameter is set to be 1, the distribution is uniform over the space of all correlation matrices; if it is larger than 1, it favors independence, expressed by zero off-diagonal terms. The larger the parameter is, the more strongly it favors independence, and thus, the more strongly it regularizes the correlation

in question.

The graphics in figure A8 below show the results of fitting the joint aggregation model (the "mu and tau" model) with an LKJ prior with concentration 1 (the default used in the previous sections), as well as 3 (moderate regularization) and 6 (strong regularization). In this case the stronger priors have no impact on the posterior hypermeans for TOT or ITT. While this is initially surprising, it reflects a small empirical correlation between the baseline and treatment effect.

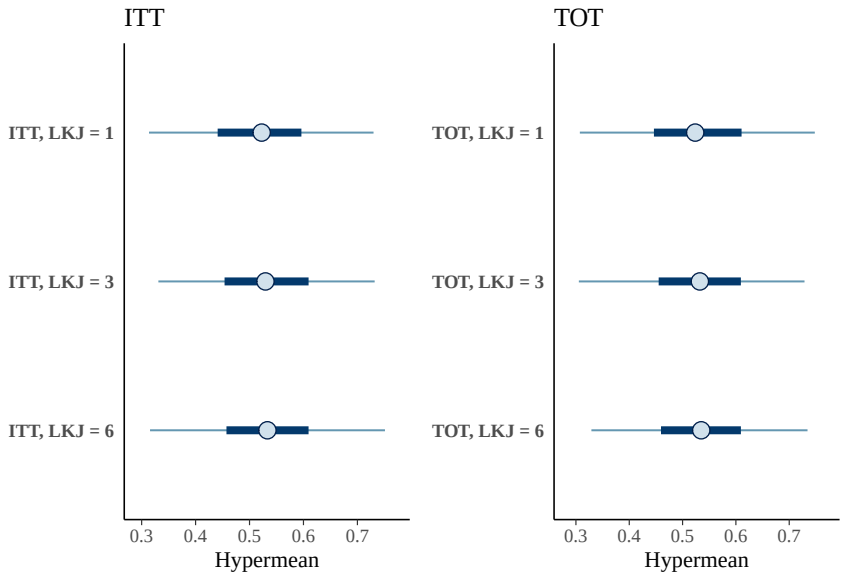


Figure A8: Mu-tau model with different LKJ priors

These results do not necessarily mean baseline levels of learning do not matter for targeted instruction to be effective. Rather it is possible that the set of studies included are all cases with relatively low baseline learning. Thus, if low baseline learning is a critical condition whereby targeted instruction is needed and effective, for all studies this condition might be met, hence positive effects across the board.

H.2 Evidence Aggregation Robustness Checks

In this section we conduct additional robustness checks that help us understand how the inference on the average effects is constructed from the sample of studies at hand, and assess whether the analysis in this paper is vulnerable to classical publication bias.

We first conduct Leave-One-Out analysis in order to understand the robustness of our main results to omitting any of the studies. This is especially a concern when we have a small number of results in a given literature, as is the case typically for aggregation of RCTs (see for example Meager 2019, where the same robustness check is presented for the microcredit RCT aggregation exercise). We take as our main analytical result of interest the average treatment effect, either in terms of ITT or TOT, across all the studies in the data set. That is, we examine the posterior distribution of the hypermean from the Bayesian hierarchical models and display its sensitivity to leaving out each of the studies in turn. These "Leave one out" sensitivity results are shown for the ITT estimates both in the Rubin (1981) partial pooling model and for the model conditioning on the implementation delivery model in figure A9, where the study indicated in the row label is the study omitted for that run of the model.

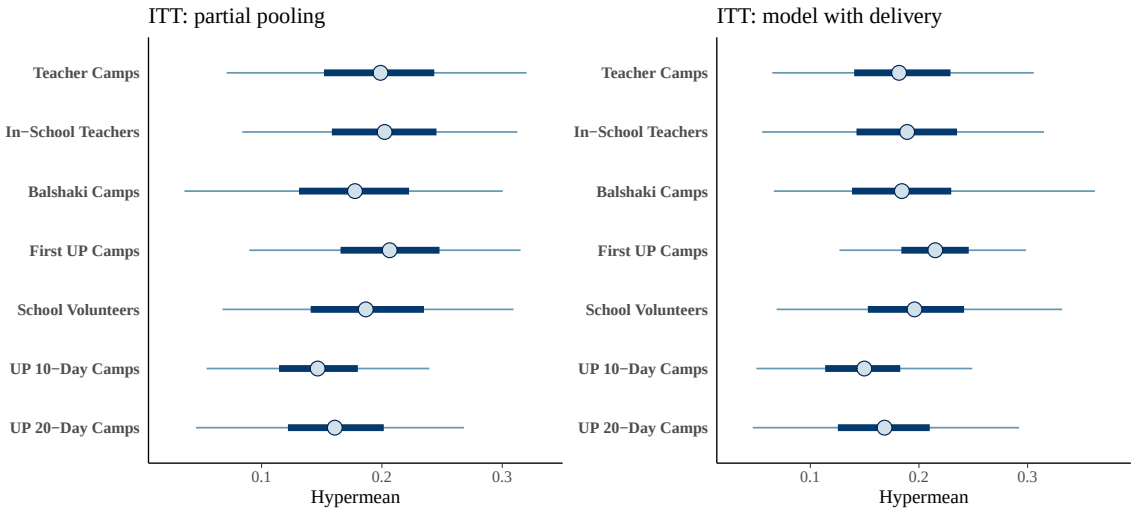


Figure A9: ITT Leave One Out analysis

The results above show relatively little variation in the posterior distribution of the hypermean when any given study is omitted, with the slight possible exception of the three UP camps estimates. These three studies each seem to exert more influence than the other studies, although they run in different directions – dropping the first UP camps tends to increase the hypermean, while dropping the 10- or 20-day camps tends to decrease the hypermean. However, in all cases there is substantial overlap in the posterior intervals with the general results, and even for the UP camps study omission the posterior mean of the hypermean is well within the central 50% credible interval of the other posteriors. This shows relatively strong robustness of the ITT results overall.

Figure A10 shows the sensitivity results of leaving out studies in turn for the TOT estimates both in the Rubin (1981) partial pooling model and for the model conditioning on implementation delivery.

The graphs clearly show little variation in the posterior distribution of the hypermean for

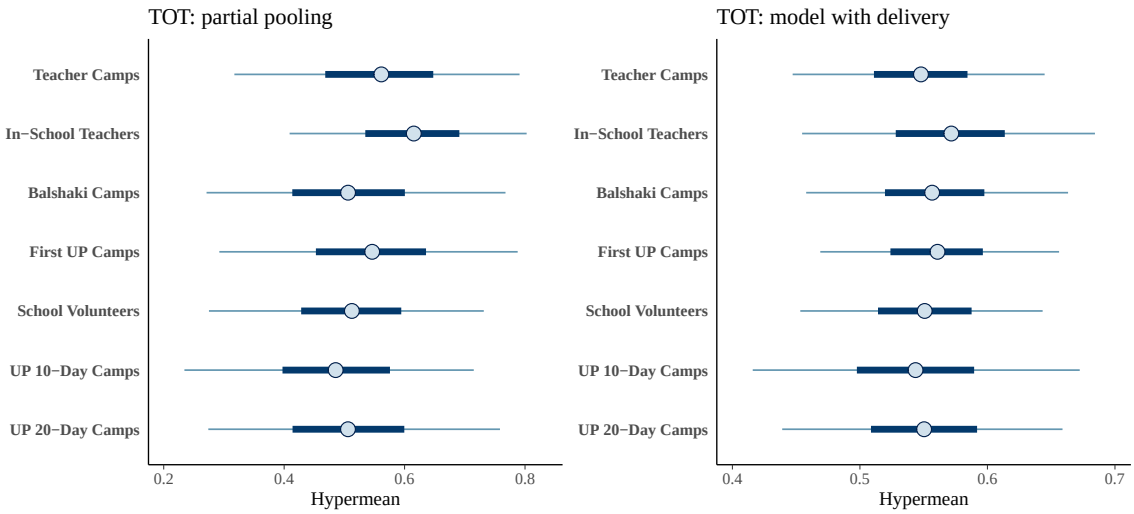


Figure A10: TOT Leave One Out analysis

either the classical Rubin partial pooling model or the meta-regressive model conditioning on delivery. The slight exception is the effect of leaving out the In-School Teachers study in the Rubin model, which has a somewhat more pronounced effect on the hypermean, but this is not present in our preferred meta-regressive specification conditioning on delivery model. This shows the strong robustness of the TOT results to omitting any of the studies, and demonstrates that our insights about the important role of implementation factors are not based on any single study but rather borne out across the literature as a whole.